

Monochrome CRT Monitor

- Design Documentation -

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1. Why This is a Terrible Idea

2. Equipment Needed

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4. Signal Spec

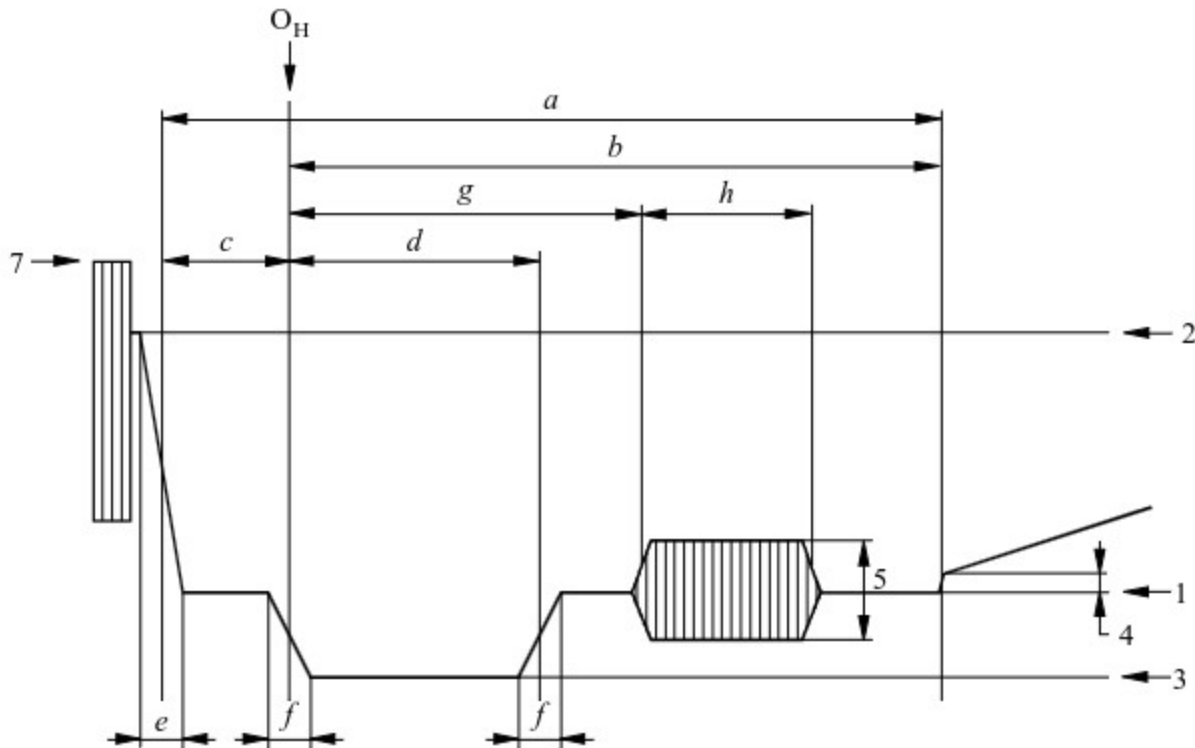


Figure 1.4 – The signal used by PAL and NTSC systems

1. Blanking level
2. Peak white-level
3. Synchronising level
4. Difference between black and blanking levels
5. Peak-to-peak value of burst
6. Something about colors
7. Peak level including chrominance signal

H - nominal line period = 64uS for PAL: Amount of time it takes the electron beam to draw a horizontal line across the screen, reset its position and prepare to draw the next line.

a - line blanking interval = 12uS for PAL:

Amount of time it takes the electron beam to fly from right to left for the next line.

d - synchronising pulse = 4.7uS for PAL: The timing signal that tells the electron beam exactly when to start drawing each new line of video and when to start a new frame.

b - Interval between time datum (OH) and back edge of line - blanking pulse = 10.5uS for PAL:

Interval from the timing reference point to the back edge of the blanking pulse. Tells you where the blanking ends relative to the sync reference.

c - front porch = $1.5\mu\text{s}$ for PAL:

The small gap between end of active video and start of sync pulse. Exists so the signal has time to settle to blanking level before the sync pulse starts.

e - Build-up time (10-90%) of the edges of the line-blanking pulse = $0.3\mu\text{s}$ for PAL:

How fast the blanking pulse switches on and off.

f - Build-up time (10-90%) of the edges of the line-sync pulses = $0.2\mu\text{s}$ for PAL:

How fast the sync pulse itself switches

Active video = $52\mu\text{s}$, derived from H minus a

PAL has no pedestal, meaning the black level is equal to the blanking level.

The signal(s) that CRTs run off of is “encoded” (or more so sandwiched together) by throwing everything into the same wire. Separation of the signals is done by the TV/monitor and is helped by the different behaviour and voltage levels that the signals exhibit:

1) 0mV - sync tip: The sync separator can tell sync signals apart by the simple fact that they sit below every other signal, nothing else ever drops this low.

2) 300mV - blanking level: When voltage is at 300mV the beam is off. No picture. This is the floor of the visible signal. The transition from sync up to 300mV is the back porch (the beam is still off but preparing to draw the next line).

3) 300mV - black level (PAL): In PAL, black level and blanking are at the same voltage. A pixel that is pure black is exactly at 300mV. In NTSC there's a small gap between blanking and black (called a pedestal), PAL doesn't have that.

4) 1000mV - peak white: Maximum brightness. A pixel that is pure white drives the signal to 1000mV.

Everything between 300mV and 1000mV is picture.

- Interlacing:

Interlacing tackles the problem of expensive bandwidth and screen flickering. The human eye perceives flicker up to $\sim 46\text{Hz}$, meanwhile the 625 lines are being ran at 25fps.

The solution to this is splitting each frame in two fields. Field 1 would render all the odd numbered lines up to 625, while field 2 all the even numbered ones up to 624.

After the first field has been rendered, the beam moves diagonally from line 625 but it can't return to the same starting point as field 1 since line 2 sits in between line 1 and line 3. The beam returns half a line width lower and starts drawing out lines 2, 4, 6, etc.

The beam takes time to get from the bottom corner of the screen back up to the top corner, and on top of that it needs to be off otherwise there would be a diagonal line over the image. It takes 25 lines worth of time per field for the beam to go back, there are 50 lines lost in total per frame.

- Resolution:

The resolution of a CRT TV/monitor is determined by the bandwidth of the video amplifier, however, the main limiting factor of this is actually the electron beam spot size, which on a typical black and white tube is 0.3-0.5mm, giving ~600-1000 distinguishable spots physically. Going beyond that causes the spots to overlap, effectively leading to zero additional improvement in resolution.

The formula for this is:

$$\text{pixels} = 2 \times \text{bandwidth (Hz)} \times \text{active line time (s)}$$

$$2 \times 5000000 \times 0.000052 = 520\text{p}$$

$$2 \times 10000000 \times 0.000052 = 1040\text{p}$$

$$2 \times 15000000 \times 0.000052 = 1560\text{p}$$

$$2 \times 50000000 \times 0.000052 = 5200\text{p}$$

As mentioned before, active line time is derived from nominal line period minus line blanking interval. So $H - a = 64\mu\text{s} - 12\mu\text{s} = 52\mu\text{s} = 0.000052\text{s}$

Increasing the output of the HV supply would accelerate the electrons faster, resulting in a narrower beam spot size and allowing for more individual pixels to be displayed. The problem is that too high a voltage would burn a hole into the phosphor screen – this can be fixed by moving the beam faster (increasing horizontal deflection frequency, but this comes with its own set of problems). The bandwidth of the video amplifier would also need to be increased for the expected number of pixels as well as the higher H deflection frequency, and on top of that your display is now incompatible with every signal in existence.

However, with color CRTs this is not the case – they use a shadow mask, a thin sheet of metal full of tiny holes, every electron gun has a pre-determined spot where it can hit the phosphor screen. Without it, the colors would just bleed into each other and hit the wrong spots, turning the image into a blurry, rainbow mess.

5. Sync

The synchronization signal is a timing signal. The TV has no way of knowing where the beam is at any given moment, the oscillators just run and hope they're in the right place. The sync pulses tell the oscillators exactly when to start each new line. The oscillators fire the reset based on the sync pulses so they can start each new line at exactly the right place and time. Without it the picture would roll and tear.

Now, how do we pull the sync information out of the entire signal? The answer to that question is an IC called "LM1881". The sync signal is the lowest part of the signal, sitting below the blank/black and peak white levels – the IC takes advantage of that. The incoming signal is AC-coupled by a capacitor on the input, so no DC can get through. On top of that, the sync tips are clamped to an internal voltage reference of $\sim 1.5V$, in other words, the entire waveform is shifted so the bottom of the sync pulses always line up at the same exact point, which is necessary so the AC coupling doesn't cause the waveform to drift around. This gets fed through a comparator with a threshold set at 70mV higher, so 1.57V, whose output is a clean digital signal, only high during sync.

This IC has three other outputs that need to be explained, vertical sync, odd/even field, and burst gate:

1. Vertical sync – The IC uses an internal integrator to distinguish the narrow H pulses (4.7uS) from the wide V pulses. It does so using an RC integrator, short/fast pulses hardly charge the capacitor while longer pulses are able to charge the capacitor much higher. Two comparators watch the level constantly checking if the integrator voltage has crossed their thresholds – this is how they tell apart V sync pulses from H pulses. The output of the two comparators is then cleaned up using flip-flops, an OR gate and a divide-by-8 so a pulse of defined length starting at vertical sync detection at 50Hz appears on pin 3. The resistor value is picked using a graph from the datasheet while the capacitor is fixed at 100nF

2. Odd/Even field – pin 7 identifies odd/even fields, useful for applications where you need to know what field you're on, irrelevant to a display only device.

3. Burst gate – pin 5 outputs a burst gate pulse for chroma subcarrier recovery, which is not needed for monochrome CRTs.

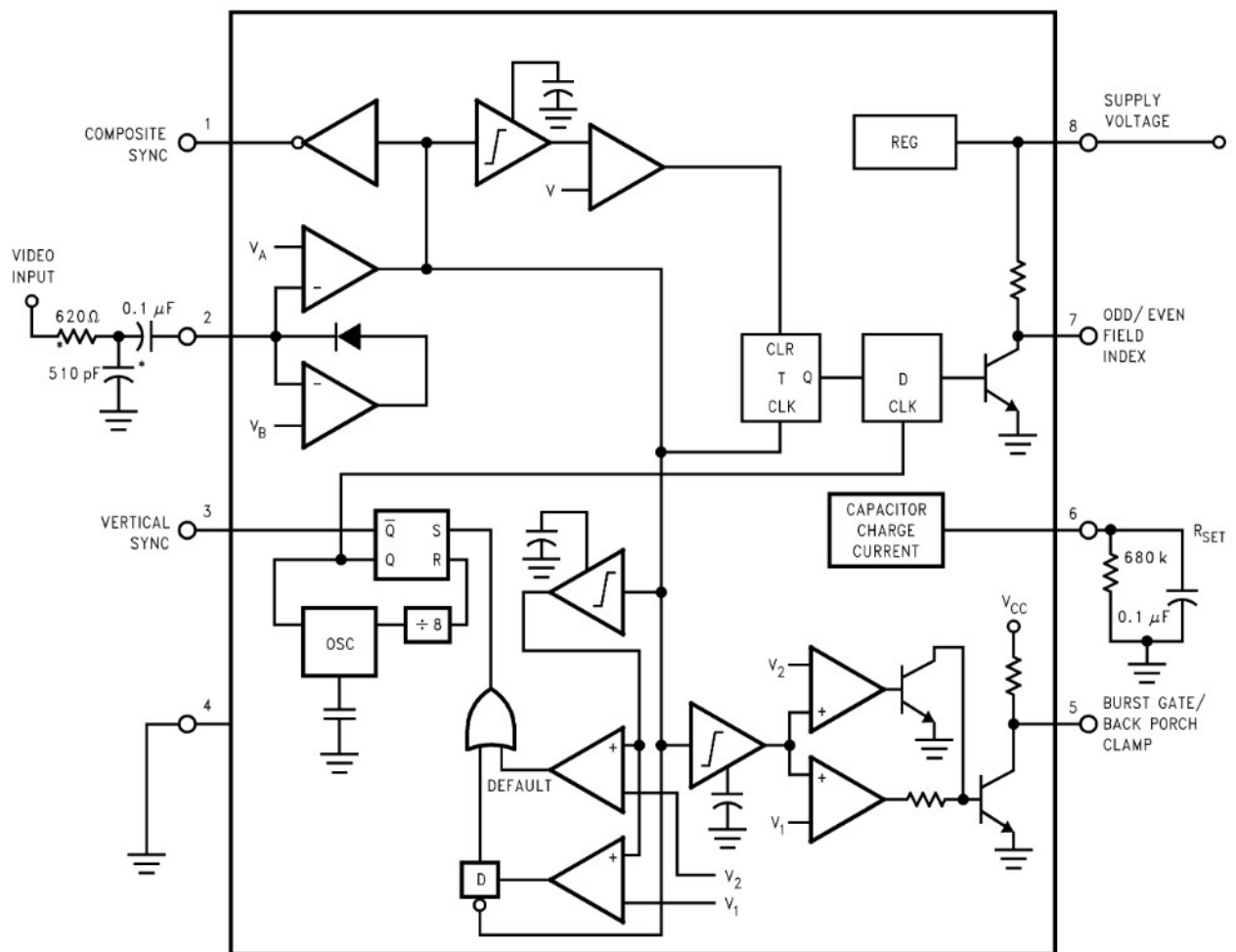


Figure 1.5 – Internal block diagram of the LM1881

- Pin 1 - Composite sync output. This is where the synchronisation signal for the horizontal deflection comes from.
- Pin 2 - Composite video input. This is where the entire signal is fed.
- Pin 3 - Vertical sync output. This is where the synchronisation signal for the vertical deflection comes from.
- Pin 4 - Ground.
- Pin 5 - Burst or back porch timing output.
- Pin 6 - Charge current external resistor.
- Pin 7 - Odd and even field output.
- Pin 8 - Supply voltage.

R_{set} sets the integrator charge current and internal oscillator frequency, value selected from datasheet graphs for PAL.

So with everything taken into consideration:

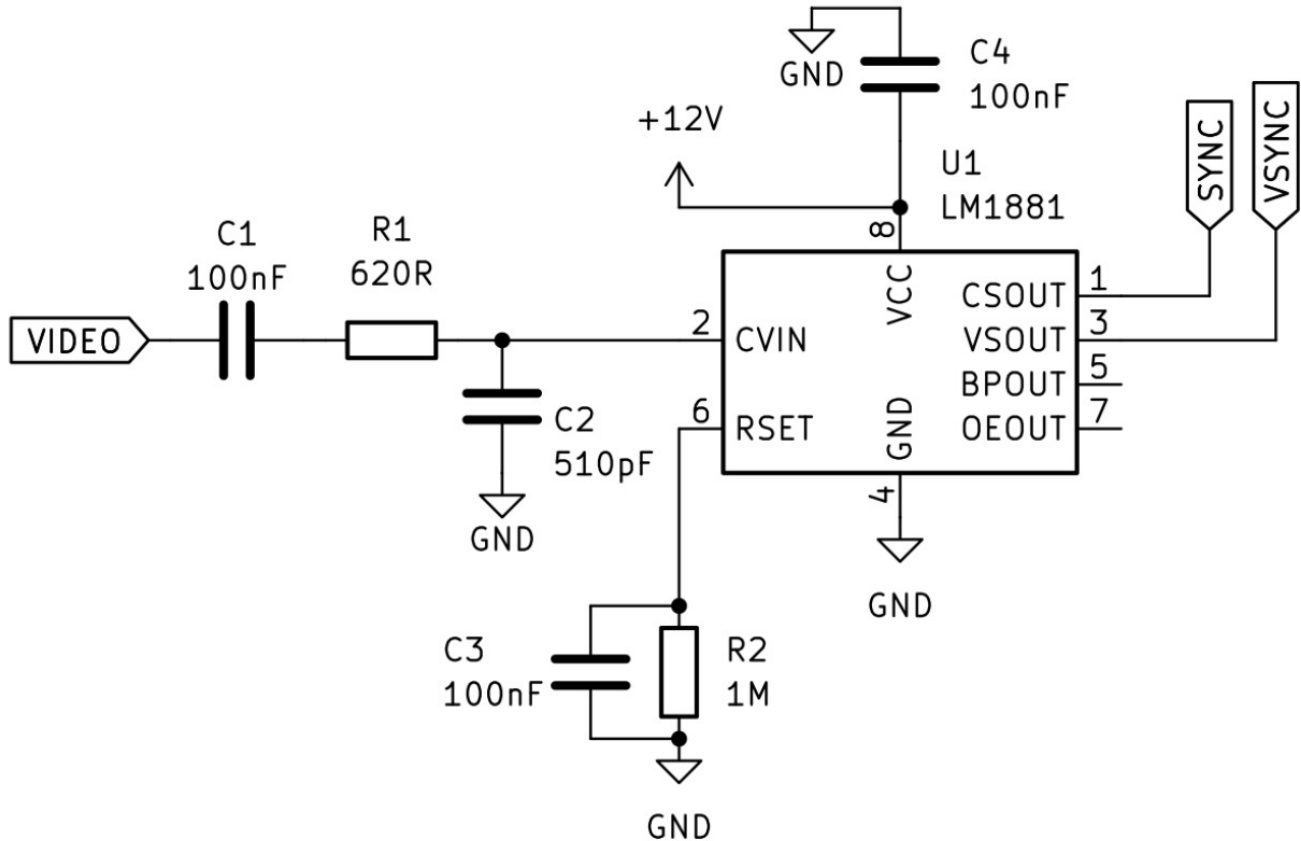


Figure 2.5 – The complete sync separation circuit

1. AC coupling cap:

The video signal first passes through the previously mentioned AC coupling cap with a value of 100nF, blocking DC from being fed into the IC.

2. Low-pass filter:

The video signal then passes through a low-pass filter comprised of a 620Ω resistor and a 510pF:

$$f_{\text{cutoff}} = \frac{1}{2\pi RC}$$

$$f_{\text{cutoff}} = \frac{1}{2 \times \pi \times 620 \times 0.00000000051} \approx 500 \text{ kHz}$$

The values used set the cutoff at around 500kHz, which is the sweetspot of filtering out the color subcarrier and other high frequency noise without smoothing the sync pulses to the point where the IC cannot tell them apart anymore.

3. R_{set} resistor:

According to the datasheet, the vertical sync delay time must be longer than the vertical sync period of 64μS, meaning that R_{set} must be larger than 680k but still small enough to allow the integrator to reach V_1 (the trip point) with a higher limit of ~2M taking into account leakage currents. A value of 1M was chosen.

6. How to Get 25kV Without Dying

There are various power converter topologies, i will go in depth on only one of them, and that is the flyback converter. A flyback converter, like any other power converter, works by cycling the switch at a specific frequency and utilising some form of feedback to adjust the duty cycle in order to regulate the output voltage. As with other types of converters, flyback converters can use either voltage mode regulation or current mode regulation.

1. Voltage mode: The error signal is compared with the internal oscillator's sawtooth waveform to control the ON time of the switch.

- Advantages:

- the simplicity of a single feedback path
- low output impedance from the power stage (the output varies less based on the load)
- good noise margins (because of the internally generated ramp)

- Disadvantages:

- the need for careful loop compensation (because of the 2-pole LC output filter)
- slow loop response (especially with input variations)
- the need for separate current-limiting circuitry for the switch transistor(s)

2. Current mode: The sawtooth is replaced by the ramping current in the primary/inductor, which is compared with the error signal. The internal oscillator is now used to initiate each conduction cycle and terminate it once the current ramp crosses the error signal.

- Advantages:

- rapid response to input changes
- inherent pulse-by-pulse current regulation on the primary
- improved phase margin in the outer voltage-feedback loop
- the ability to parallel the outputs of several identical converters

- Disadvantages:

- the greater difficulty of managing two nested feedback loops (mitigated by widely separating their characteristic frequencies)
- more affected by load changes because the fast loop tends toward a constant current output
- more susceptible to noise, especially at low load, and to resonances (caused by the PWM's dependency on the current derived ramp).
- premature termination of the switch's ON-state caused by the leading-edge current spike (from parasitic capacitances and diode recovery effects)
- instabilities and subharmonic effects at high duty cycle

Bottom line, both methods compare the output voltage to an internal reference voltage to generate an error signal. The difference stands in what they compare the error signal to and what they do with that output:

- Voltage mode – compares the error signal with an internally generated sawtooth waveform which is used to adjust the duty cycle.
- Current mode – compares the error signal to the current ramp from the primary/inductor, with the internal oscillator now used to initiate conduction and terminate it once the two signals cross.

Both approaches are viable as their disadvantages can be mitigated via various tricks: The slow response to input changes of voltage-mode controllers can be fixed by adding an input feedforward signal to the sawtooth ramp, as for the slow loop response the solution is using a higher switching frequency. When it comes to current-mode controllers, leading-edge blanking can be used to fix the issue of the switch-ON current spike, a small addition to the circuit that tells the controller to ignore the first fractions of a second at power-on, and slope compensation can be used to restore stability at higher duty cycles.

Flyback (and other) converters diversify even more as they can work either in discontinuous conduction mode (DCM) or continuous conduction mode (CCM):

1. DCM – The inductor/transformer fully discharges before the next cycle.
2. CCM – The inductor/transformer doesn't fully discharge before the next cycle.

Where flyback converters stand apart is the fact that they use a transformer (or more of a tightly coupled inductor) instead of a single inductor for energy storage.

A flyback transformer functions based off a two phase operation:

1. ON phase:

During this phase, the current in the primary ramps up (according to $V_{in} = L_{pri} \frac{di_{pri}}{dt}$), the input energy is going entirely into the magnetic field of the transformer's core. The diode on the secondary is reverse biased so nothing is conducting yet.

2. OFF phase:

As the supply is cut off, the magnetic field collapses and flips polarity on the secondary, effectively dumping all of its energy into the output capacitor and the load.

The core serves as a means of transfer/temporary storage for the energy, which is what makes it more of a coupled inductor rather than a transformer. Moreover, the flyback transformers used in CRTs have a gap to raise the saturation current and to provide a more linear inductance, without it, the core would saturate faster and the winding would become a near dead short to whatever is driving it.

The diode on the secondary is there to enforce the two-phase operation, ensuring nothing can conduct while the MOSFET is still on and at the same time rectifying the output.

The output voltage is multiplied by the turns ratio:

$$N = \frac{N_{pri}}{N_{sec}} \text{ for step-down converters and } N = \frac{N_{sec}}{N_{pri}} \text{ for step-up converters}$$

With the energy ramping down according to:

$$V_{out} = L_{sec} \times \frac{di_{sec}}{dt}$$

The output voltage can be estimated with (for DCM):

$$V_{out} = V_{in} \times D \times \sqrt{\frac{R \times T_s}{2 \times L_m}} \times \left(\frac{N_{sec}}{N_{pri}} \right)$$

Where:

D - switch duty cycle (ratio between 0 and 1)

R - equivalent load resistance (in ohms)

T_s - switching period, equal to $1/f_s$ (frequency in seconds)

L_m - transformer magnetizing inductance, usually measured on the primary side (in henries)

As for CCM:

$$V_{out} = V_{in} \frac{N_{sec}}{N_{pri}} \frac{t_{on}}{t_{off}} = V_{in} \frac{N_{sec}}{N_{pri}} \frac{D}{1-D}$$

A flyback transformer allows for multiple secondary windings, and thus multiple output voltages, however the regulating feedback can only be taken from a single winding, which leaves the other windings only regulated via transformer coupling, meaning they will vary a lot more based on the load. Now, a little about the flyback I'm using:

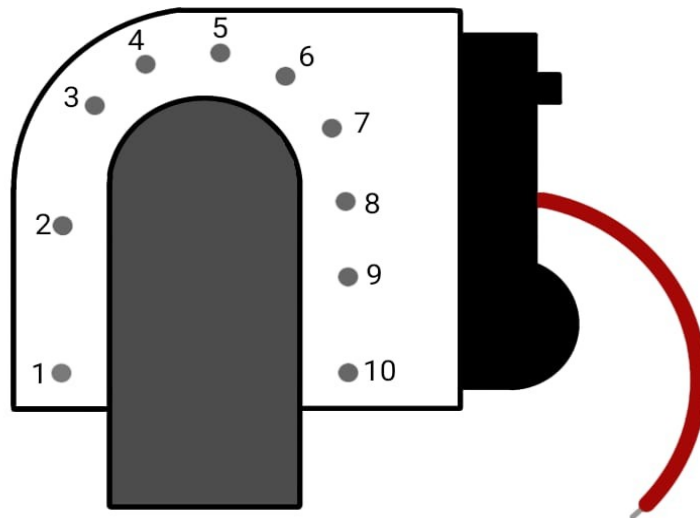


Figure 1.6 – Pinout diagram of a CRT flyback transformer

It is the original transformer from the CRT I salvaged for this project. To the surprise of no one, the inscriptions on it were of no use to identifying its pinout but I was lucky enough for the PCB to have the pins labeled:

1. “C1” – primary
2. “C2” – primary center tap
3. “C3” – primary
4. “+9” – positive 9V auxiliary winding
5. “+15V” – positive 15V auxiliary winding
6. “-15V” – negative 15V auxiliary winding
7. “NC”
8. “ABL” – automatic brightness limiter
9. “HER” – electron gun heating element supply
10. “180V” – positive 180V auxiliary winding

I managed to find online the anode voltage of a similar CRT tube which “should not exceed 27kV” and measured the inductance of the primary winding (290uH).

The only problem left is that of the supply voltage. Budget CRT sets from that era run the HOT at 80-130V, with the search result that gave me the anode voltage also mentioning 125V B+. Meaning that the turns ratio of this transformer is realistically sitting somewhere in between 1:190 and 1:250.

Assuming a turns ratio of 1:200, in order to reach 25kV with ~30% headroom and a 70% max duty cycle (CCM formula used due to the absence of the load resistance value, bench tweaking required):

$$V_{in} = V_{out} \times \frac{N_{pri}}{N_{sec}} \times \frac{1-D}{D}$$

$$V_{in} = 25000 \times \left(\frac{1}{200} \right) \times \left(\frac{0.3}{0.7} \right) = 125 \times 0.4286 \approx 53.6 \text{ V}$$

And for a 50% duty cycle:

$$V_{in} = 25000 \times \left(\frac{1}{200} \right) \times \left(\frac{0.5}{0.5} \right) = 125 \times 1 = 125 \text{ V}$$

In other words, the supply voltage required will be around 100-125V for a comfortable margin, however, the actual bring-up will be done via variac.

The next unknown worth tackling is the frequency. Normally, the flyback transformer is driven by the HOT at the same time with the horizontal deflection coil. This has the advantage of being cheaper, which in the 1940s was understandable, but the main drawback is that the transformer can only run at whatever frequency the horizontal deflection is running at (in the case of PAL being 15.625kHz), with little to no room left for regulation or improvements. The only way to find out if there is a frequency better suited to drive the transformer is by driving the transformer. For this, I chose this simple circuit:

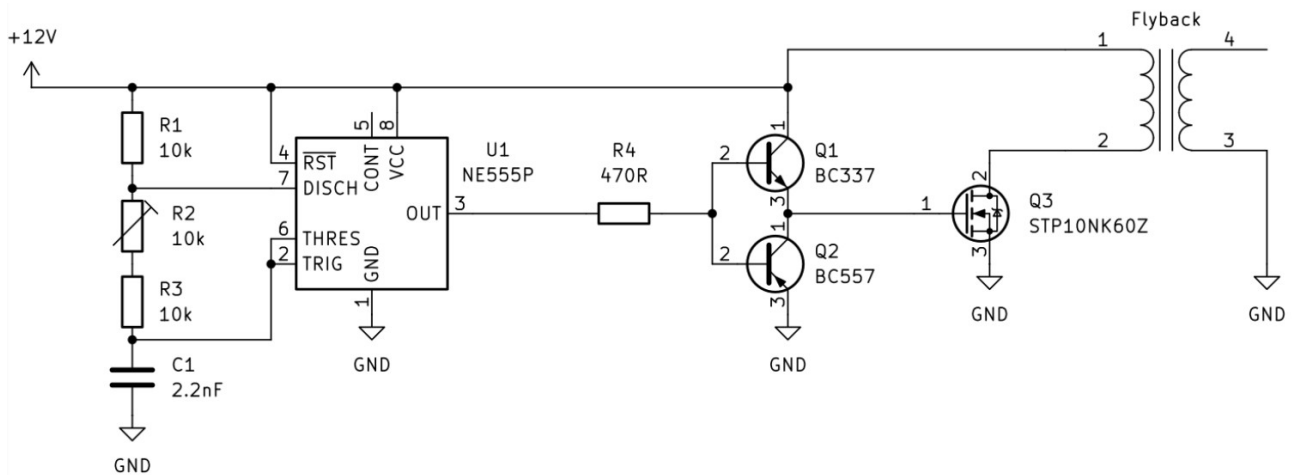


Figure 2.6 – Transformer characterization circuit

The values of R1, R2, R3 and C1 yield a frequency range of ~11-57kHz, plenty for this purpose. I used a totem pole configuration to ensure the MOSFET is properly turning on and off.

I did not bother with supply filtering or a snubber on the MOSFET since I only need this to run for about 5 minutes. I probed between various pins while sweeping R2 to see what frequency squeezes the highest voltage out of this transformer – the sweetspot turns out to be 20kHz:

1. 180V – NC: 360V
2. HER – NC: 4V

3. ABL – NC: 1.7kV

And the NC pin is apparently a common return path for the auxiliary windings (and possibly the HV secondary as well). Turns out NC doesn't always mean "no connect".

For the actual driver, I have chosen the UC3843 PWM current-mode controller because that's what I had on hand and best understood by now paired with the tighter regulation of current-mode control.

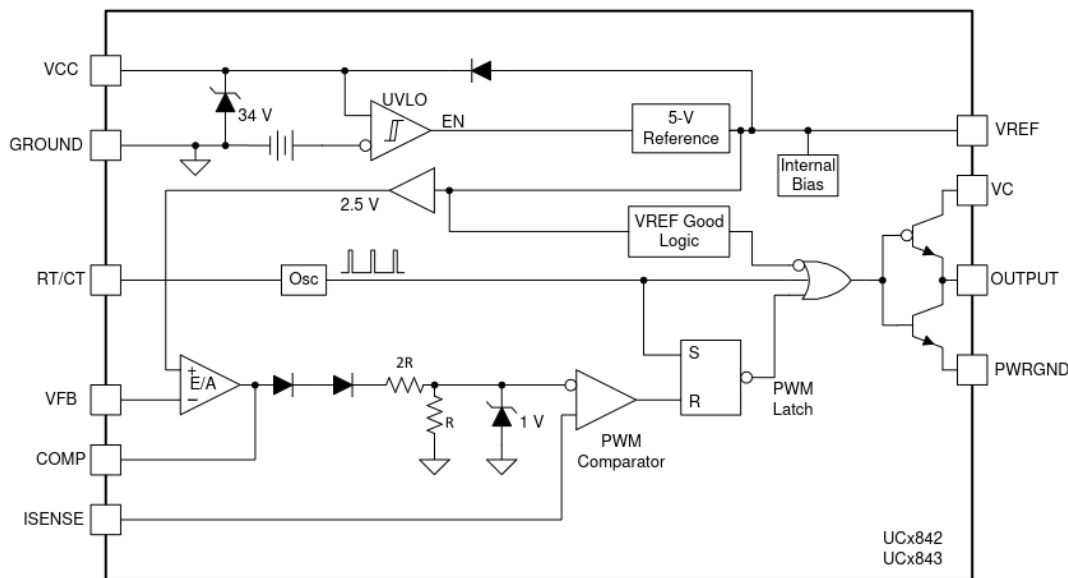


Figure 3.6 – Internal block diagram of the UCx843

Pin 1 - COMP: Error amplifier compensation pin. The error amplifier output can be modified by connecting external compensation components to this pin. The error amp is internally current limited so 0% duty cycle can be forced by shorting this pin to GND. Without the compensation components, the error amp starts oscillating – the RC network adds a pole and a zero to slow down the loop response enough to regulate tightly while keeping it stable.

Pin 2 - VFB: Inverting input of the error amplifier. The HV feedback divider connects here. The reference voltage used for comparison is 2.5V. The difference is what influences the duty cycle – if the input is below 2.5V, it increases, if above 2.5V, it decreases. An increase in load causes the HV output to sag and the reference voltage to go below 2.5V, which in turn causes the IC to increase duty cycle, transferring more energy per cycle and effectively increasing the HV back to its target, and vice versa. The formula for this is none other than the voltage divider formula:

$$V_{\text{out}} = \frac{V_{\text{in}} \times R_2}{R_1 + R_2}$$

Pin 3 - ISENSE: Primary-side current sense pin. The current sense resistor goes here. This signal is used by the PWM to terminate the output conduction. A voltage ramp can be applied to this pin to run the device in a voltage-mode control configuration.

$$I_{\text{sense}} = \frac{V_{\text{sense}}}{R_{\text{cs}}}$$

V_{sense} is 1V for this IC and R_{cs} is whatever is in series with the primary to allow I_{peak} plus headroom.

Pin 4 - RT/CT: Fixed frequency oscillator set point. The timing resistor connects between this pin and VREF (pin 8), while the timing capacitor is connected between this pin and ground to set the switching frequency. The capacitor's leads must be kept as short and as direct to ground as possible, a separate ground trace for it is preferred. The resistor's value must be kept above 5k, while the capacitor can be anywhere in between 1-100nF. The output frequency can be estimated using this formula:

$$f_{\text{osc}} = \frac{1.72}{R_t \times C_t}$$

Pin 5 - GND: Analog ground. Both signal and power ground can be connected here on packages with no dedicated power ground pin.

Pin 6 - OUTPUT: The output drive for the external MOSFET which includes an internal totem-pole configuration allowing for direct drive. Peak currents of up to 1A can be sourced and sunk from this pin. The output is actively held low when VCC is below the turn-on threshold of this IC (8.4V).

Pin 7 - VCC: Supply input. A 0.1uF capacitor connected to ground through a short and direct path is required on this pin. An additional bypass cap with a value at least 10 times greater than the FET's gate capacitance is also needed.

Pin 8 - VREF: 5V precision reference output. Can be used as a stable 5V reference for other parts of the circuit. Its primary function is biasing the RT resistor.

With all of that taken into consideration, this is what I could come up with:

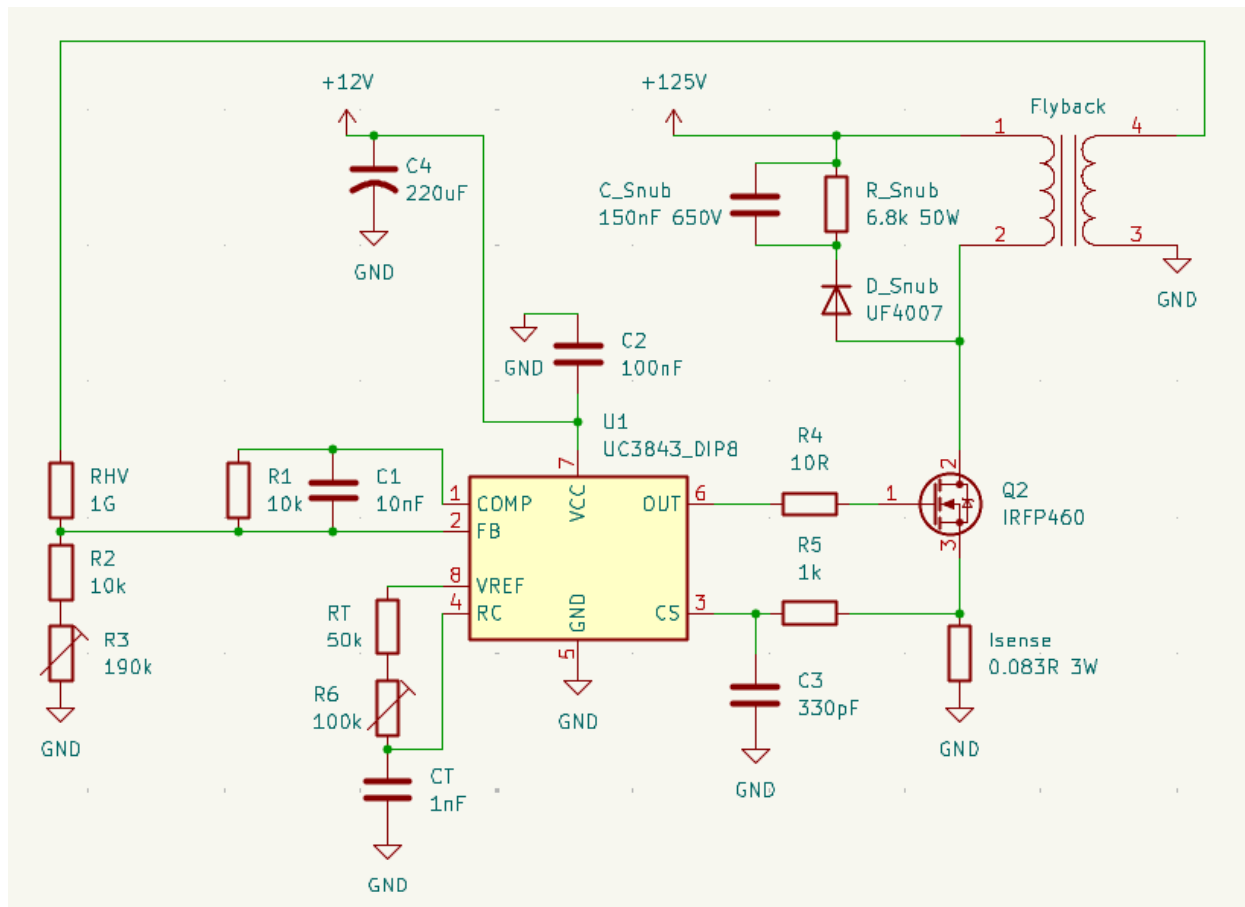


Figure 4.6 – The completed flyback driver

I have decided to bake in adjustability at every corner so as to avoid swapping out components.

1. Frequency:

With respect to the ranges on values a 1nF capacitor was chosen.

$$f_{osc} = \frac{1.72}{50000 \times 0.000000001} = \frac{1.72}{0.00005} = 34400 \text{ Hz}$$

$$f_{osc} = \frac{1.72}{150000 \times 0.000000001} = \frac{1.72}{0.00015} = 11466 \text{ Hz}$$

With a 50k fixed resistor and a 100k trimmer the final frequency range comes out to 11.4-34.4kHz.

2. Voltage divider:

The target voltage is 25kV and we need to get that down to 2.5V without loading the supply too much:

$$V_{out} = \frac{25000 \text{ V} \times 100 \text{ k} \Omega}{1 \text{ G} \Omega + 100 \text{ k} \Omega} = 2.5 \text{ V}$$

$$V_{out} = \frac{25000 \text{ V} \times 10 \text{ k} \Omega}{1 \text{ G} \Omega + 10 \text{ k} \Omega} = 0.25 \text{ V}$$

$$V_{out} = \frac{25000 \text{ V} \times 200 \text{ k} \Omega}{1 \text{ G} \Omega + 200 \text{ k} \Omega} = 4.999 \text{ V} \approx 5 \text{ V}$$

A 10k fixed resistor and a 200k trimmer were chosen.

Now to make sure the supply isn't being overloaded:

$$I = \frac{25 \text{ kV}}{1 \text{ G}} = 25 \mu \text{ A}$$

$$P = 25000 \text{ V} \times 25 \mu \text{ A} \approx 0.625 \text{ W}$$

So the top resistor needs to be 1G, at least 2W and with a voltage rating of 30-40kV – not exactly an easy find (or a cheap one) but the feedback divider is the last place where I'd cut corners.

3. Current sense resistor:

The current sense resistor must be sized appropriately to allow the peak current to flow through the primary each cycle with some headroom ensuring it doesn't terminate early.

$$I_{\text{peak}} = \frac{V_{\text{in}} \times D \times T_s}{L_n}$$

$$I_{\text{peak}} = \frac{125 \times 0.5 \times 0.00005}{0.00029} \approx 10.77 \text{ A}$$

So I_{peak} comes out to nearly 11A, let's call it 12A for headroom. Rearranging the I_{sense} formula for resistance:

$$R_{\text{cs}} = \frac{V_{\text{isense}}}{I_{\text{peak}}}$$

$$R_{\text{cs}} = \frac{1 \text{ V}}{12 \text{ A}} \approx 0.083 \Omega$$

Call it 0.082Ω as that's the closest value available to me.

We now have the value of the current sense resistor but we must size it appropriately so it doesn't catch fire:

$$I_{\text{rms}} = I_{\text{peak}} \times \sqrt{\frac{D}{3}}$$

$$I_{\text{rms}} = 10.77 \times \sqrt{\frac{0.5}{3}} \approx 4.4 \text{ A}$$

So now for power:

$$P = I_{\text{rms}}^2 \times R_{\text{cs}} = 4.4^2 \times 0.082 = 1.59 \text{ W}$$

Call it 3W for, you guessed it, headroom.

4. RCD snubber:

No transformer is perfect, flyback transformers are no exception - in an ideal circuit, the magnetic field couples completely to the secondary and all is well, but in reality, there is a small leakage inductance, a small part of the magnetic field that doesn't couple to the secondary, and which, upon shutoff, is dumped back into the primary as a positive voltage spike. The fix is a snubber network, either an RC across the winding or, preferably, an RCD network (a diode in series with a parallel RC).

This one is a little more special as no amount of math is gonna save you from bench tweaking, you only get a better starting point for the values. And unfortunately you can't over-compensate too much or it just steals energy from your driver.

First things first, we need to know the reflected voltage:

$$V_{\text{drain}} = V_{\text{in}} + \left(\frac{V_{\text{out}} \times N_{\text{sec}}}{N_{\text{pri}}} \right) + V_{\text{leakage}}$$

$$V_{\text{drain}} = 125 + \left(\frac{25000 \times 1}{200} \right) + 50$$

$$V_{\text{drain}} = 125 + 125 + 50$$

$$V_{\text{drain}} = 300 \text{ V}$$

I pulled V_{leakage} straight out of thin air. Also, call it 400V for headroom.

Now, onto the snubber resistor:

$$R_{\text{sn}} = \frac{V_{\text{sn}}^2}{\frac{1}{2} L_{\text{lk1}} i_{\text{peak}}^2 \frac{V_{\text{sn}}}{V_{\text{sn}} - n V_{\text{o}}} f_s}$$

V_{o} is the output voltage, n is the turns ratio, and V_{sn} is the reflected voltage.

Leakage inductance is 1-5% of the primary's inductance (290uH) – using the 5% figure for headroom that comes out to about 14.5uH. The main source I relied on for understanding the RCD snubber (Ridley Engineering) advises against using rules of thumb for the leakage inductance and instead measuring it by shorting the secondary (or secondaries) and measuring the primary, but that requires an LCR meter I don't own.

$$R_{\text{sn}} = \frac{400^2}{\frac{1}{2} \times 0.0000145 \times 10.77^2 \frac{400}{400 - 0.005 \times 25000} 20000} \approx 6.8 \text{ k}$$

Yeah, that was a lot. It's actually closer to 6540.233839 for anyone interested.

Now we don't want this resistor catching fire either:

$$P = \frac{V_{\text{sn}}^2}{R_{\text{sn}}}$$

$$P = \frac{400^2}{6800}$$

$$P \approx 24 \text{ W}$$

Call it 50W for the beloved headroom, though to be noted that if the leakage inductance is 1% the power dissipated comes out to ~5W.

Thankfully, the snubber capacitor is nowhere near as damning:

$$C_{\text{sn}} = \frac{V_{\text{sn}}}{\Delta V_{\text{sn}} \times R_{\text{sn}} \times f_s}$$

$$C_{\text{sn}} = \frac{400}{20 \times 6800 \times 20000} \approx 147 \text{ nF}$$

Nearest standard value is 150nF.

ΔV_{sn} - maximum allowable ripple on the clamp voltage – basically how much the snubber capacitor voltage swings between cycles which is 5-10% of V_{sn} . I chose the 5% figure and that's 20V.

The diode is nothing special, it only needs to meet two conditions: recover fast enough and withstand the reverse voltage, the UF4007 does both.

5. Compensation network:

The values for the compensation network on pin 1 were chosen based on standard UC3843 application notes and circuits, setting a pole at somewhere around 1.6kHz – conservative enough to be stable for a flyback driving a near-open-circuit anode load. As usual, fine tuning is left to bringup.

6. Gate resistor:

The gate resistor is chosen so as to balance switching losses with electromagnetic interference (EMI). A higher value slows down the turn-on and turn-off of the MOSFET, which decreases EMI but increases switching losses. The typical range for this resistor is 10-47 ohms.

Since this isn't going into any CRT to be mass produced and the only EMI concern is regarding the surrounding circuitry, not much thought went into this one, I figure 10 ohms is as good of a value as any, with bench tweaking always being an option.

SAFETY

I cannot possibly stress enough how important safety measures can be when working with something like this. While 25kV from a single transistor flyback driver isn't as deadly as 25kV from something like a ZVS driver, it can still be enough to kill you. Here are the safety precautions I took while working on this circuit:

1. Don't touch anything while it's running.
2. Don't touch anything while it's running.
3. Properly discharge the transformer, and eventually, the CRT as well as any other capacitors capable of delivering shocks if you ever need to modify or replace something.
4. Don't touch anything while it's running.
5. Ensure the HV output of the flyback cannot short to anything you care about.
6. Maintain proper physical distance from the HV output. The dielectric breakdown of air is $\sim 3\text{kv/mm}$ (varies based on the composition and properties of the surrounding air as well as electrode geometry, sharp points can do far better than that as they concentrate the electric field). At 25kV, best case scenario (round electrodes) that's an $\sim 8\text{mm}$ gap.

You either learn to be safe or this hobby will beat it into you.

7. Horizontal Deflection

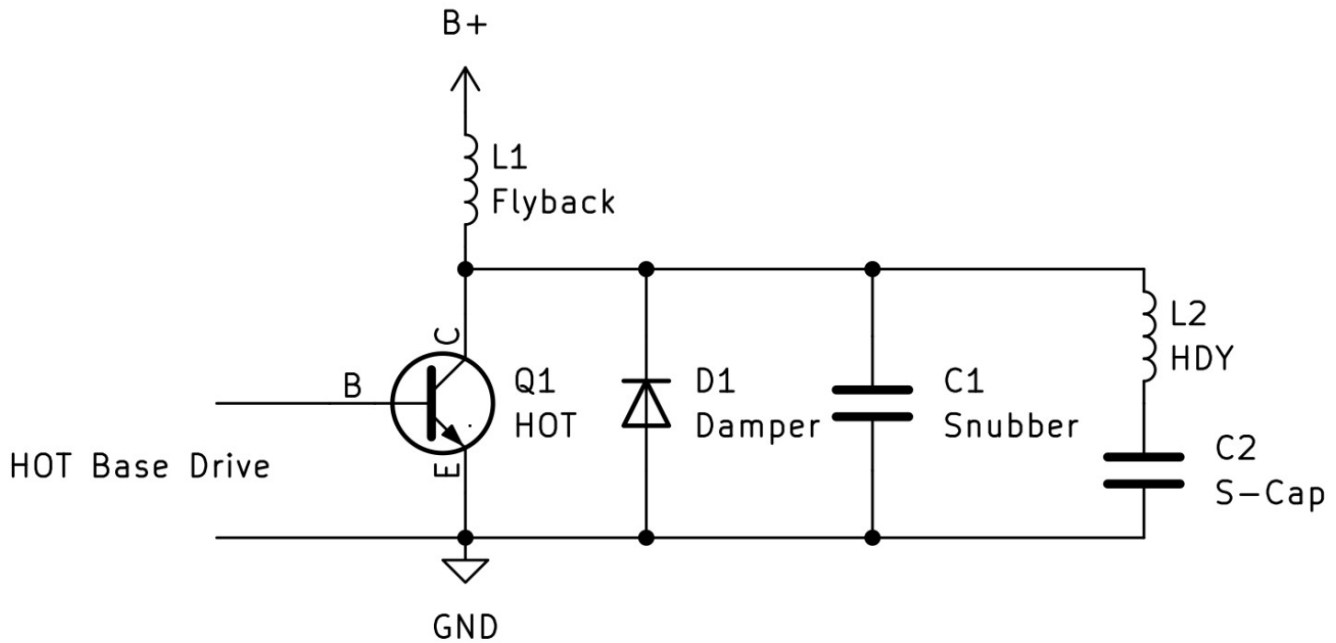


Figure 1.7 – Basic horizontal deflection output stage

- Basic horizontal deflection circuit operation:



Figure 2.7 – Horizontal deflection yoke (HDY) current waveform

The circuit starts off at something called retrace – the beam has just finished a complete horizontal line (52uS) and needs to jump to the other side to begin a new one (12uS) – that jump is called retrace.

1. At the end of a scan, current is flowing through the primary of the flyback transformer to the HOT. When Q1 turns off (must be done so in a controlled manner), the flyback period begins – this is the part of the waveform where it falls from $+I_{peak}$ all the way down to $-I_{peak}$. An inductor opposes changes in current, so upon Q1's shutoff, the current cannot be instantly cut out so it diverts into the snubber capacitor, C1. A resonant circuit is now formed by the transformer's primary and C1, so that voltage on C1 rises as current goes down. At its peak, this voltage will be $\sim 1-1.5kV$.

2. C1 begins discharging in reverse through L1 (the primary of the flyback transformer) until its voltage reaches 0.

3. Normally, the oscillation under C1 and L1 would continue below 0V into the negative but D1, with the anode at ground and the cathode at node A (where L1, Q1, D1, C1 and HDY meet), immediately clamps down any negative voltage as it becomes forward biased – this is the part of the waveform that rises from -Ipeak to 0A.

Steps 1 through 3 have successfully accomplished the function of reversing the current in L1 (and the deflection yoke). The full flyback (and yoke) current is now flowing through the damper diode, D1.

4. When a new scan begins, the still forward biased diode carries the entirety of the current from the yoke and flyback. The nearly constant supply voltage causes a linear ramp of current through the diode since it's still negative and decreasing in magnitude.

5. At around mid-scan, the current changes polarity from minus to plus cutting off the damper diode and causing the prematurely turned on HOT to pick up the current. Current is now being drawn from the supply – this is the part of the waveform that rises from 0A up to +Ipeak.

6. During the second half of the scan, the current through the HOT rises up linearly thanks to the supply voltage being nearly constant across the flyback's primary.

7. As the end of the scan approaches, the HOT turns off and the cycle repeats. Also, the transistor has a "storage" time of a few μ S that needs to be accounted for so as to allow time for the remaining charge carriers to clear out before it properly turns off, but this is an issue for its driving circuitry.

- **Correction Circuits:** Since most CRTs aren't flat and the electron gun is close, the image would suffer linearity problems and pincushion distortion without any corrections applied.

1. **S-Correction:** The electron beam covers more space at the same speed around the edges of the screen compared to the center (because geometry), so a capacitor is added in series with the yoke – the S correction cap/S-cap. The capacitor together with the yoke forms a series LC circuit with a resonant frequency below the line frequency and adds the effect of making the peaks of the current ramp smoother which reduces the scan speed at the edges of the screen. Without it the picture will be stretched at the sides. Multiscan monitors require circuitry that selects the S-cap value based on the signal input (PAL, VGA, etc).

2. **N/S and E/W Correction (Tube Dependent):** The beam has to travel further to reach the corners compared to the edges, which leads to pincushion/barrel distortion - the corners bow inwards or outwards. N/S (north/south) is vertical pincushion – on smaller tubes (sub 100 degrees) it can be fixed by yoke design, sometimes requiring a small high frequency injection into the V deflection waveform. E/W (east/west) is horizontal pincushion – it causes the left and right edges to bow inwards and requires an active electronic fix by modulating the horizontal scan width with a parabola shaped signal derived from the vertical deflection, in other words, wider scan in the middle and narrower at the top and bottom.

3. Horizontal Linearity Correction: Without any resistance the constant supply voltage creates a linear ramp across the yoke. But once resistance is added, as the current increases the voltage drop increases as well ($V = IR$). This slows down the ramp towards +Ipeak, making it asymmetrical. The easiest way of fixing this is by adding a negative resistor, but Mouser doesn't stock those so we have to get clever by using an inductor with a magnetic core in series with the yoke (the linearity coil) – at first, the current is low, the inductance stays high, but as the current increases the core (eventually) saturates and inductance drops to near-zero. The linearity coil slows down the ramp at its beginning, it doesn't fix the effect of the voltage drop towards the positive peak, it makes it symmetrical. This coil is usually an adjustable inductor far smaller in value compared to the yoke.

- HOT Base Drive: The HOT base drive is comprised of two sections: an oscillator generating a square wave at 15.625kHz (for PAL) locked to sync and a driver transistor to amplify that signal for the HOT's base current draw and drive it through a transformer. The driver transistor is usually a medium power NPN switching the transformer's primary and in turn feeding the HOT's base via the secondary. Apart from supplying current to the base, the transformer also generates a negative going pulse upon shutoff to help clear out remaining charge carriers and cut down on the HOT's storage time. In between the HOT and transformer there's usually a resistor that sets the base current and affects storage time - this can be used for optimizing specific HOTs.

- PLL and the CD4046 IC:

A phase-locked loop (PLL) is a feedback system that locks an oscillator to an external signal by permanently correcting its frequency so it matches. It is comprised of a phase comparator, a loop filter and a VCO.

The phase comparator takes two inputs, the H sync pulse from the LM1881 and the VCO output, and produces an error signal proportional to the phase difference between them. The error signal is filtered by the loop filter (low-pass RC network) to remove high frequency noise which leaves a slowly varying DC control voltage. The loop settles when the phase difference is zero meaning the VCO is running at exactly the reference frequency. Now, the RC filter is fine for only the XOR comparator, the edge triggered comparator requires a different approach as you'll see later.

The VCO is running freely at around the sync pulse frequency and the loop pulls it into lock. Once the lock is achieved any deviations in the RC timing components are continuously corrected, holding the picture horizontally still.

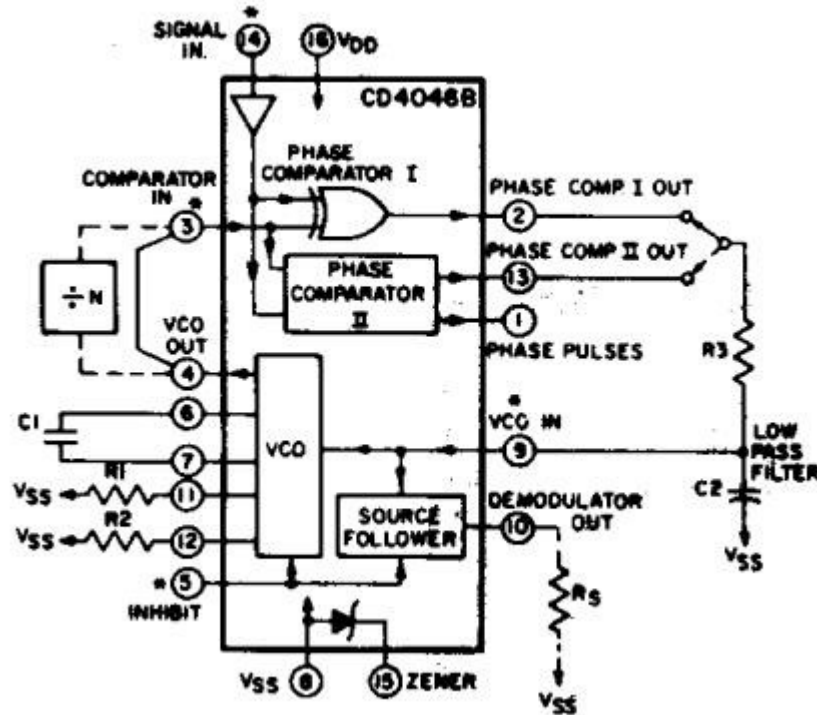


Figure 3.7 – Internal block diagram of the CD4046 IC

Designed in the 70s, documented in the 70s, still documented like it's the 70s, and somehow the only good option, the CD4046 IC contains both the VCO and two phase comparators. Pin 2 is an XOR comparator (PC1), simpler but causes a shift in phase after locking and is sensitive to duty cycle, pin 13 (PC2) is an edge-triggered comparator, produces no phase shifts at lock and ignores the duty cycle, preferred for this application.

R1, R2 (pin 11 and pin 12, must be above 10k and below 1M) and C1 (pin 6 and 7, must be above 50pF at 10V) set the VCO frequency with R1 setting the maximum frequency, C1 the center frequency and R2 the minimum frequency. The loop filter connects between the VCO input (pin 9) and the phase comparator output (pin 13). The loop bandwidth and transient response is set by a series RC to ground – too wide and noise pulls the VCO around, too narrow and the loop is slow to lock and correct shifts, but, contrary to the datasheet, you can ignore that since a simple RC circuit is not enough for the edge-triggered comparator, instead, an active integrator comprised of an op-amp and a couple passives is required here.

The VCO output is pin 4 which goes to the rest of the base drive circuitry.

So with all of that taken into account, this is what I was able to come up with:

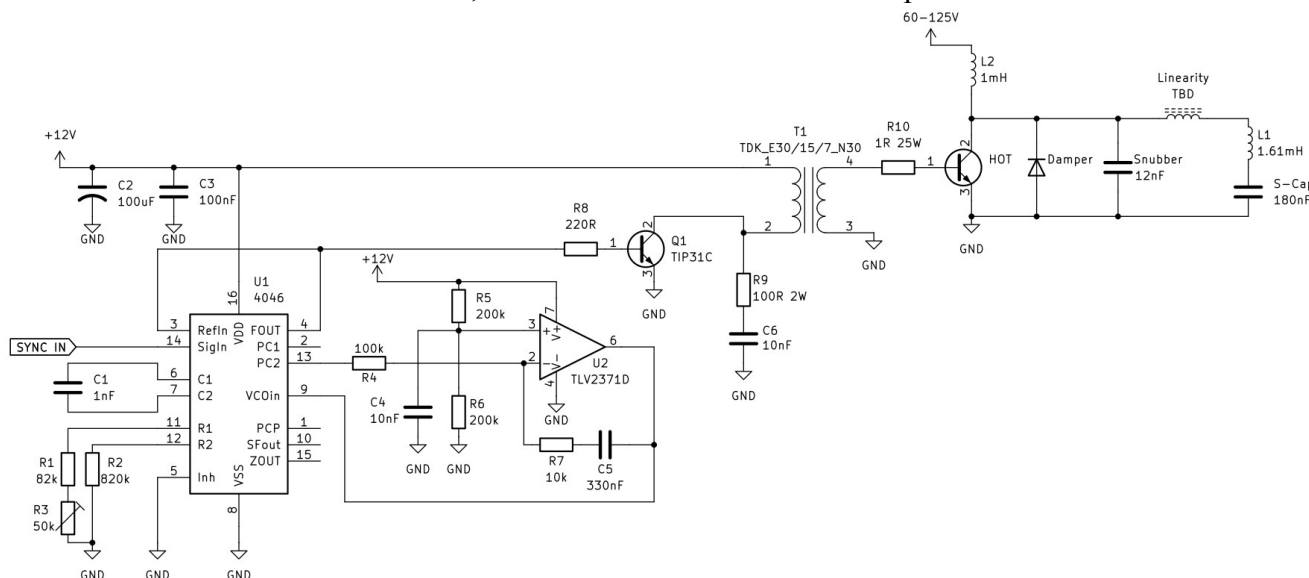


Figure 4.7 – The (nearly) complete horizontal deflection

Just as with the HV supply, I chose adjustability over fixed values where I saw fit so as to avoid swapping out components.

1. CD4046 Timing Components: This part was fun, there are no formulas to calculate anything here as they'd lie to you, instead, you have to look at figure 5 and 6 from its datasheet and figure it out yourself:

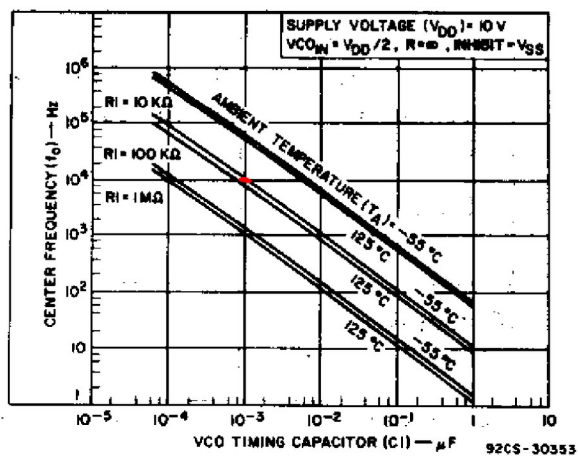


Fig. 5 – Center frequency as a function of $C1$ and $R1$ for ambient temperatures of -55°C to 125°C .

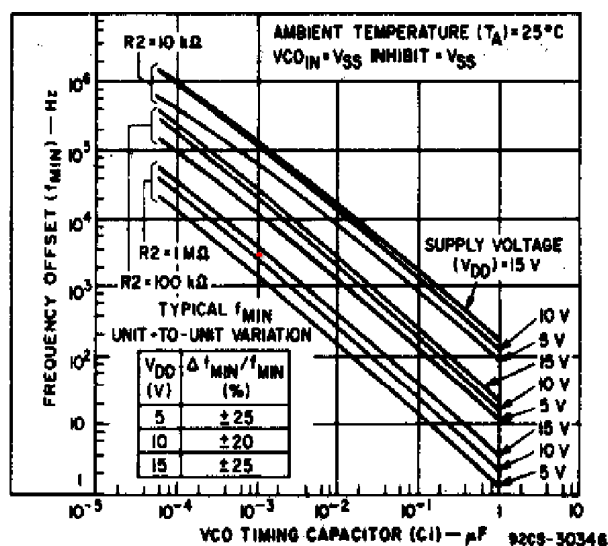


Figure 5.7 – CD4046 datasheet graphs used for picking $R1$, $C1$ and $R2$ values

Yes, the frequency is supply dependent, I used the 10V line in the graphs since that's closest to my supply voltage of 12V and picked 1nF. I think you can see why there's a trimmer on this one. I landed on 100k for $R1$ (82k fixed + a 50k trimmer) and 820k for $R2$ (which just needs to be considerably below your target frequency, not that critical).

2. Active integrator loop filter:

As previously mentioned, a simple RC circuit won't be enough for the edge-triggered comparator, an active integrator is needed.

The loop filter is rather simple, comprised of a single op-amp and a couple passives, its job is to take the raw error signal of the phase comparator and output a slowly varying DC voltage which is used to tell the VCO how much to correct its frequency. The op-amp has to be able to swing rail-to-rail, with 0V giving f_{MIN} and 12V f_{MAX} and 15.625kHz landing somewhere in between this range.

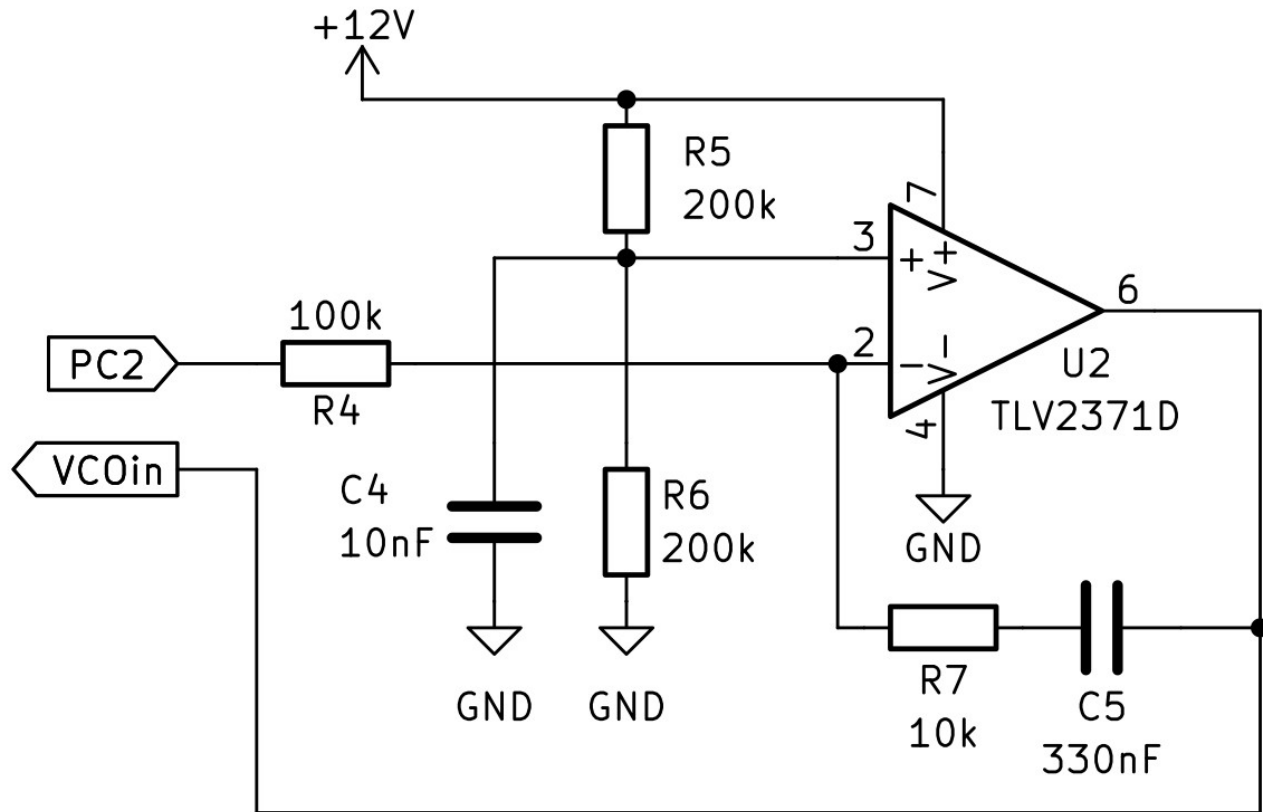


Figure 6.7 – The active integrator loop filter

While getting into the math behind the components, a few parameters need to be defined:

- KP – Phase detector gain. How many volts the phase comparator outputs per radian of phase error between the two inputs. This is fixed by the supply voltage:

$$KP = \frac{V_{DD}}{4\pi}$$

$$KP = \frac{12\text{ V}}{4\pi} = 0.954929$$

- KN – Feedback divider ratio. It's $1/N$ where N is the amount of times the VCO output is divided before feeding it back to the IC, I'm not dividing so $N=1$, $KN=1$.

- KV – VCO gain. How many radians per second the VCO frequency shifts per volt of control voltage. Depends on f_{MAX} and supply voltage:

$$KV = \frac{2\pi \times f_{MAX}}{VDD}$$

$$KV = \frac{2\pi \times 17000}{12V}$$

$$KV = 8901.179185$$

- ω_n – Natural frequency of the loop. How fast the loop responds to disturbances.

$$\omega_n = 2\pi \times \frac{f_{in}}{200}$$

$$\omega_n = 2\pi \times \frac{15625}{200}$$

$$\omega_n = 490.8738521 \text{ rad/s}$$

- ζ – Damping factor. How the loop behaves when correcting an error. Below 1 it overshoots, at 1 it's critically damped, while above 1 it's sluggish but stable. I went with 0.8.

Now onto the passive components:

R4 was picked to be 100k because I like the orange color on the resistor. For finding C5:

$$C = \frac{KP \times KV}{N \times \omega_n^2 \times R}$$

$$C = \frac{0.954929 \times 8901.179185}{1 \times 490.8738521^2 \times 100000} \approx 352 \text{ nF}$$

Nearest standard value is 330nF.

As for R7:

$$R_7 = 2 \times \zeta \times \frac{\sqrt{N \times R_1}}{\sqrt{KP \times KV \times C}}$$

$$R_7 = 2 \times 0.8 \times \frac{\sqrt{1 \times 100000}}{\sqrt{0.954929 \times 8901.179185 \times 0.00000033}}$$

$$R_7 = 9553.3062$$

Nearest standard value is 10k.

We now need to check if the squiggly line is still 0.8:

$$\zeta = R_7 \times \frac{\sqrt{KP \times KV \times C}}{2 \times \sqrt{N \times R_4}}$$

$$\zeta = 10000 \times \frac{\sqrt{0.954929 \times 8901.179185 \times 0.00000033}}{2 \times \sqrt{1 \times 100000}}$$

$$\zeta = 0.8374064225$$

So the final values come out to:

$$R4 = 100k$$

$$R7 = 10k$$

$$C5 = 330nF$$

The voltage divider on the non-inverting input needs to be twice the size of R4 and step down the supply voltage to half, while C4 is just a decoupling cap. The TLV2371 op-amp was chosen as it's able to swing rail-to-rail and has a GBW of 3MHz.

3. Base drive transformer:

The transformer driving the HOT needs to meet two requirements:

- Provide enough current through the base resistor to fully saturate the transistor.
- Introduce a negative going pulse of enough magnitude upon shut-off so as to clear the charge carriers and turn the HOT off fast enough.

We need to pick a proper core fit for 15.625kHz and large enough that it doesn't saturate during the 52uS time period of active video with a sane amount of turns. The suitable core material is N30, the size isn't quite as straightforward though:

$$N_{pri} = V \times \frac{t}{B_{sat} \times A_e}$$

Where:

V – primary voltage (12V)

t = scan period (52uS/0.000052s)

B_{sat} - Saturation flux density. The maximum magnetic flux density the core material can handle before reaching saturation and essentially turning the winding into a short circuit. Measured in Teslas or mT (300mT).

A_e - effective cross-sectional area of the core (60mm²)

Plugging those in:

$$N_{pri} = 12 \times \frac{0.000052}{0.3 \times 0.000006} \approx 35$$

Call it 40 turns for headroom.

So in order to not drive the core into saturation and allow the magic smoke to escape an N30 core with a cross sectional area of 60mm² is needed – I picked the TDK E30/15/7 N30 core.

Now onto sizing the secondary:

As per the FJL6920TU datasheet, for an 11A collector current we need a 2.75A base current while introducing a negative going pulse upon shutoff of 4A for a 3uS storage time.

So the secondary needs to shove 2.75A of current into the base and then create a pulse that's able to sink 4A all through a 1Ω resistor.

Starting off, we need the voltage output of the secondary:

$$V_{sec} = (I_{base} \times R_{base}) + V_{BE}$$

$$V_{sec} = (2.75 \times 1) + 1.5 = 4.25 \text{ V}$$

So for 4.25V:

$$N_{\text{sec}} = N_{\text{pri}} \times \frac{V_{\text{sec}}}{V_{\text{pri}}}$$

$$N_{\text{sec}} = 40 \times \frac{4.25}{12} = 14 \text{ turns}$$

So now that we know the secondary can properly turn on the HOT, we have to make sure it can properly turn it off. We do that by confirming that the transformer's magnetizing inductance stores enough energy to actually sink 4A:

Magnetizing inductance of the primary is given by:

$$L = A_L \times N^2$$

Where A_L for this specific core is given by the manufacturer and is 3100nH.

$$L = 0.0000031 \times 40^2$$

$$L = 0.00496 \text{ H}$$

We can find the energy stored in the inductance at a peak primary current of 1A with:

$$E = \frac{1}{2} \times L \times I^2$$

$$E = \frac{1}{2} \times 0.00496 \times 1^2$$

$$E = 0.00248 \text{ J}$$

We now have everything needed to check if 2.48mJ is enough to sink 4A:

$$E = I^2 \times R \times t$$

$$E = 4^2 \times 1 \times 0.000003 = 0.000048 \text{ J}$$

So in order to sink 4A we need 48uJ with 2.48mJ available.

We can size the transistor driving the transformer based on the principle of conversion of power:

Secondary delivers 2.75A at 4.25V which is 11.69W. In an ideal transformer, primary power must equal secondary power, so the current in the primary can be found using the following formula:

$$I = \frac{W}{V}$$

$$I = \frac{11.69 \text{ W}}{12 \text{ V}} \approx 1 \text{ A}$$

In order to leave headroom we can size the driver for 2A, the TIP31C transistor can do 3A. As per its datasheet, hFE minimum is 25 at $I_C = 1\text{A}$:

$$\frac{1}{25} = 40 \text{ mA}$$

A 220Ω resistor allows for 51mA.

Only thing left to do here is power sizing the base resistor. Knowing the HOT is on for 52uS and off for 12uS, we can calculate the duty cycle:

$$\text{Duty} = \frac{t_{\text{on}}}{t_{\text{on}} + t_{\text{off}}} = \frac{t_{\text{on}}}{T}$$

$$\text{Duty} = \frac{52 \mu\text{s}}{64 \mu\text{s}} = 0.8125$$

So with a duty cycle of ~81% we get:

$$P = 2.75^2 \times 1 \times 0.8125 \approx 6.15 \text{ W}$$

Call it 15W for headroom.

So to recap:

Transformer core: TDK E30/15/7 N30

Primary: 40 turns, 0.5mm enameled copper

Secondary: 14 turns, 1mm enameled copper

Driver transistor: TIP31C

Base resistor: 1Ω , at least 15W

4. Horizontal output stage:

I have chosen a more empirical approach to this one. As you can see it is comprised of nothing but the basic circuitry lacking any kind of correction past the S-cap and the linearity coil which is still to be decided. The actual plan is first getting a raster going and only then worrying about whatever corrections it might need.

Its supply voltage is heavily dependent on the peak current required through the deflection yoke and there's really only a handful of ways to determining it:

- measure it on its original PCB (undoable in my case)
- calculate it

So let's try and calculate it:

$$V_{\text{supply}} = L \times \frac{2 \times I_{\text{peak}}}{T_s} + I_{\text{peak}} \times R$$

Rearranging for I_{peak} we get:

$$I_{\text{peak}} = \frac{V_{\text{supply}}}{\frac{2L}{T_s} + R}$$

The inductance and resistance of the yoke were measured to be 1.61mH and 2.6Ω .

T_s is simply active line time which is 52uS or 0.000052s.

With a supply voltage of 60V (which is the unimaginably good scenario) we get:

$$I_{\text{peak}} = \frac{60}{\frac{2 \times 0.00161}{0.000052} + 2.6} \approx 0.93 \text{ A}$$

Just shy of 1A.

Meanwhile a more realistic scenario of 125V gives us:

$$I_{\text{peak}} = \frac{125}{\frac{2 \times 0.00161}{0.000052} + 2.6} \approx 1.94 \text{ A}$$

Nearly 2A.

So from here on out I only have two options: either design and build a variable power supply with a range of at least 80-125V able to output 3A or do bringup using a variac with a full bridge rectifier and a beefy capacitor, find the supply voltage where the image looks just right and then design the actual supply. I prefer the latter.

Now, onto the S-correction cap:

$$C_s = \frac{T_s^2}{\pi^2 \times L}$$

$$C_s = \frac{0.000052^2}{9.87 \times 0.00161} \approx 170 \text{ nF}$$

I picked 180nF since that's the closest nominal value.

As for the snubber/retrace capacitor:

$$C_r = \frac{T_s^2}{\pi^2 \times L}$$

$$C_r = \frac{0.000012^2}{9.87 \times 0.00161} \approx 9.06 \text{ nF}$$

I went with 12nF since anything lower is going to fry the HOT while anything higher might slow down retrace too much. This is left for the bench to figure out.

Now in order to know what transistor to use we need to see what voltage it has to handle:

$$V = I_{\text{peak}} \times \sqrt{\frac{L}{C_r}}$$

Assuming worst case scenario of 125V or an I_{peak} of 2A that gives us:

$$V = 2 \times \sqrt{\frac{0.00161}{0.000000012}} \approx 732.6 \text{ V}$$

Now adding that to the supply voltage for the actual value we get 857.6V. With that taken into consideration, the FJL6920TU was chosen, however there are some issues that need to be mentioned:

V_{ceo} - the collector-emitter breakdown voltage with the base open is 1700V, which is plentiful.

V_{ces} - the collector-emitter breakdown voltage with the base shorted to the emitter is 800V, which is below the calculated 857V. However I'm sticking with this transistor because V_{ces} is the guaranteed minimum, real devices break down 20-30% higher and with negative base drive during retrace the effective breakdown is higher, and on top of all of that, 125V is the worst case scenario which is unconfirmed, I expect the actual peak current to be somewhere around 1.2-1.5A which would give a collector voltage of (roughly) 500-700V. The only other option apart from hoping for a lower supply voltage to be enough is increasing the value of the snubber capacitor which I don't want to do.

Moving on to the damper diode, it simply needs to recover fast enough and withstand the calculated collector voltage, the VS-HFA08TB120-M3 was chosen – it's an ultra fast soft recovery diode with a reverse voltage rating of 1200V.

As for L2, its purpose is to replace what would normally be the primary of the flyback transformer. There wasn't any math done here, I settled on 1mH because it's not that far from the ballpark and is the

lowest value that behaved best during simulations of the output stage, as usual, adjustment is left to bringup.

8. Vertical Deflection

The vertical deflection is comprised of two main active components:

1. Vertical Oscillator: a circuit produces a 50Hz (PAL) sawtooth wave that's synchronized to vertical sync pulses for stable frame timing. The circuit normally includes controls such as vertical hold control to adjust the frequency and get a clean lock and vertical height control which directly controls the output amplitude, thus the picture height.

2. Vertical Output Stage: The sawtooth voltage is converted to a sawtooth current to drive the deflection yoke. The amplifier functions as a current driver rather than a voltage amplifier. Feedback is commonly used to linearize the ramp and compensate for non-idealities and inductance. The output stage also has to handle retrace (flyback) in order to return the beam to the top of the screen. The vertical deflection circuit is much more simple compared to the horizontal deflection because it doesn't need to do anything else other than generate a sawtooth current at 50Hz – it's primarily an oscillator and a power amplifier.

For this I have chosen the TDA1170D IC, which contains the entire vertical deflection system in a single package:

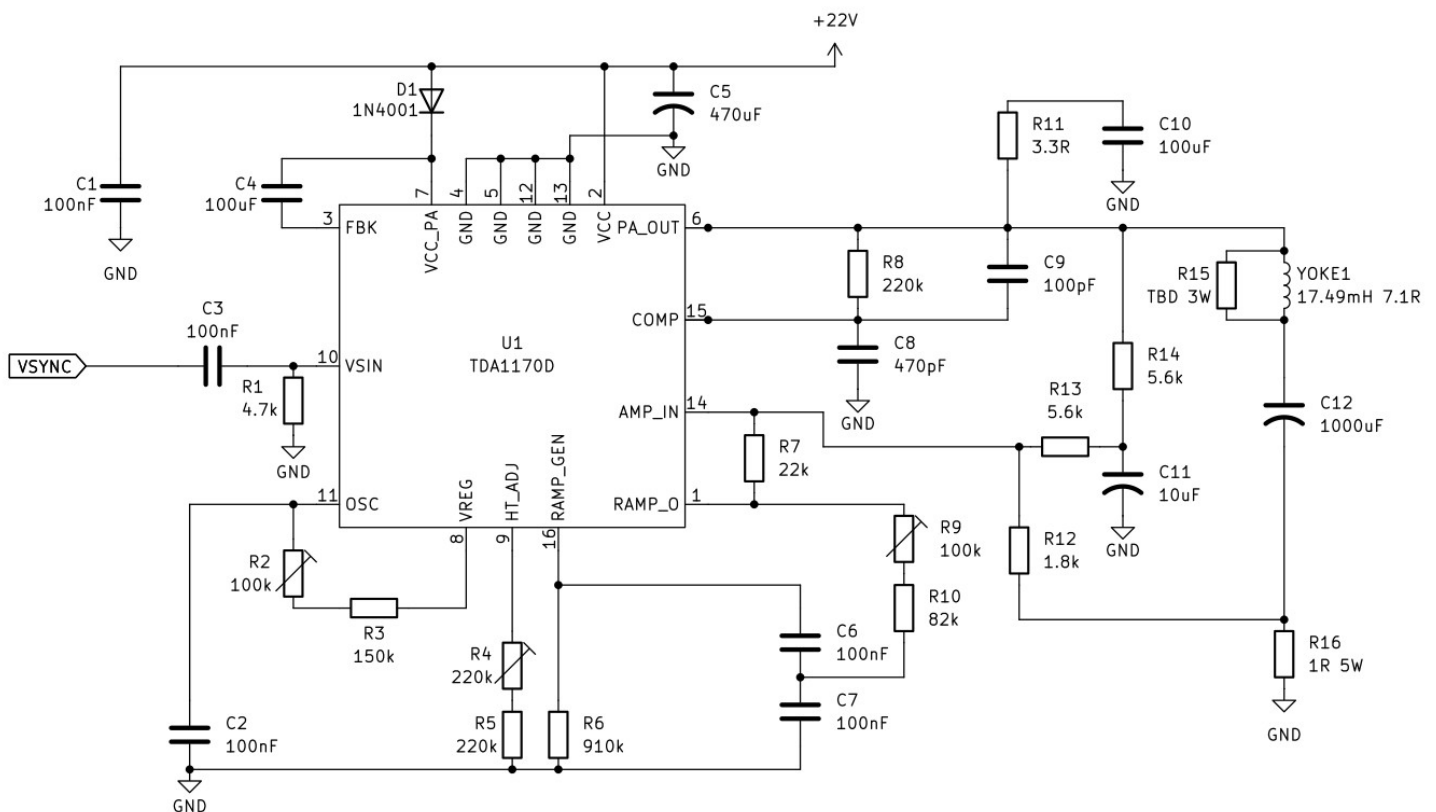


Figure 1.8 – The complete vertical deflection circuit

Not much math was done here as the datasheet just hands you the diagram and quite literally nothing else.

1. Sync Tickling:

The vertical sync signal is fed to pin 10 of the comically large square via an AC-coupling cap (C3). The capacitor is shunted to ground through R1 so that way the input isn't floating, setting the input impedance and allowing the coupling cap to charge and discharge properly between edges.

2. Oscillator:

The components that set the oscillator's timing are C2, R2 and R3, all picked from the application schematic.

3. Height:

The components used to adjust picture height are R5, setting a baseline, paired with the R4 trimmer for adjustment, also picked from the application schematic.

4. Linearity:

Just like with horizontal deflection, the further the electron beam moves away from the center the more ground it covers at the same speed for the same reason a laser pointer traces a wider gap on a wall the further you tilt it from the center (geometry and stuff).

The trimmer R9 fixes this by blending in a small S-shaped (parabolic) correction on top of the linear ramp, making the waveform rounder at the top and bottom, thus reducing the beam's speed near the extremities. Its value was taken from, you guessed it, the datasheet.

5. Compensation:

The internal amplifier driving the yoke is a feedback amplifier which requires frequency compensation in order to not oscillate and remain stable. This is accomplished using R8 and C8, and the reason they aren't baked into the chip is because they're yoke-dependent and the IC was designed to work across a range of yoke values, I picked mine from the datasheet's application circuit for the 20mH yoke.

6. HV Energy Recovery:

During flyback, the IC's internal flyback generator throws out a pulse into the yoke of 40-50V so it can snap the beam back into place in time, well above the supply voltage. Comprised of R11 and C10, their purpose is to form a path that allows some of that energy to go back to the supply instead of it dissipating entirely in the output stage, a form of snubbing if you will. The datasheet doesn't call for a specific power rating on anything here, so a 1/4W resistor it is.

7. Yoke Damping:

In order to prevent the yoke's inductance from forming an LC tank with its inter-winding capacitance, a resistor is placed in parallel with it. The value is to be determined on the bench granted there are any oscillations to begin with.

8. Current Sense Resistor:

The circuit uses R16 to measure the current through the yoke and feeds it back to the IC as a voltage so it knows when to terminate the cycle (sound familiar?). The datasheet calls for a 1W resistor but I'm using a 5W one.

9. Supply:

The IC can run on anything from 10V up to 35V. I'm going with 22V since that's what the application circuit uses for a yoke of 20mH and 10Ω , which isn't that far off from mine. With C1 and C5 used to stabilise the supply and kill any transients that might show up as well as compensate for whatever voltage sag might happen during retrace (meaning C5 has to have a low ESR).

Though it should be noted that the supply voltage alone isn't what determines whether the yoke will work properly at 22V, that is dependent on its deflection sensitivity (measured in mm/A, mm/mA or degrees/A) which tells us how much the beam moves for a given current, something that also applies to the horizontal yoke and a value that I simply cannot get from anywhere, my only option being bring-up via variac.

9. Video Amplifiers and Other Forms of Self-Harm

1. How an amplifier works:

Think of an amplifier like a valve. You rotate the valve in order to allow water to flow. An amplifier functions in a similar fashion, a small signal (your hand) controls how much current (water) can flow from the supply through a load. The small signal simply steers a power supply. Every single amplifier is some sort of implementation of the "small signal controls big current" principle.

Take a BJT common emitter stage:

- a small current fed into the base controls a larger one between the emitter and collector
- base current goes up > collector current goes up > more voltage drop across the load resistor > voltage at the collector goes down
- the inversion is inherent: input goes up, output goes down
- the ratio of the output swing to input swing is called gain

A tube functions similarly: a voltage on the grid controls current through the anode instead of a current on the base controlling the one through the collector.

2. What a pentode is and why the EF80:

A basic triode tube is comprised of three elements (hence the name): the cathode (boils off electrons), the grid (controls how many get through) and the anode (collects the electrons). The problem with triodes is their high capacitance in between the anode and grid feeding the output back into the input at higher frequencies, destroying bandwidth and leading to instability - this is negligible for audio frequencies but unusable for video.

This can be fixed by adding multiple electrodes between the grid and anode to further isolate them from one another, like a pentode, which has five elements (hence the name again):

- Cathode
- Control grid G1: controls the electron stream
- Screen grid G2: held at a fixed positive voltage to shield G1 from the anode, accelerates electrons past G1 regardless of anode voltage
- Suppressor grid G3: connected to cathode, deals with secondary emission
- Anode: collects the electrons

G2's shielding reduces the grid-to-anode capacitance to essentially nothing, which is why the EF80 can work at video frequencies of 10s of MHz.

G2's acceleration effect makes anode current nearly independent of anode voltage which is controlled almost in its entirety by G1 – this is pentode behavior. The parameter describing how well G1 voltage controls anode current is called g_m (transconductance). Higher g_m means a smaller grid voltage can produce a bigger current change, yielding a higher gain.

G3 isn't all that important past connecting it to the cathode.

3. The Characteristic Curves:

Within the EF80's datasheet a graph containing various curves can be found (which are crucial during design): The X-axis is anode voltage (V_a), Y-axis is anode current (I_a). There are multiple curves each corresponding to a specific G1 voltage while G2 is held fixed.

Each curve can be seen rising steeply at low V_a followed by an almost fully flat region - that's pentode behavior, the point where anode current stops caring about anode voltage which is now set entirely by G1. The rising region during low V_a is the tube turning on, during which the anode isn't positive enough to properly collect the electrons - operating here will cause the tube to behave like a triode which, as mentioned, will cause instability and lower bandwidth.

The spacing between the flat regions of the multiple curves (how much I_a changes per volt of G1) is gm. Tightly packed curves show a lower gm while widely spaced ones signify a higher gm.

Cutoff: past a certain threshold of negative G1 voltage, the current drops to basically zero - beam off, black level.

Key terms:

- V_a = anode voltage
- I_a = anode current
- G1 = control grid
- G2 = screen grid
- gm or S = transconductance (mA/V)

4. The Load Line:

The tube is connected in series with a resistor which is then connected to the supply, the entire drawn current goes through both of them which causes a constraint:

$$V_a = V_{cc} - (I_a \times R_L)$$

That's a straight line which can be found via two ways:

If I_a is zero then V_a is equal to supply voltage, full supply appears at the anode since there is no current being drawn meaning no voltage drop across R_L

If V_a is zero then I_a is equal to supply voltage divided by R_L . All the voltage vanishes into nothingness across the load resistor and there's none left for the tube

That line is drawn over the myriad of other characteristic lines – where it crosses a specific G1 line is where the circuit actually functions for that specific G1 voltage, which is also the point that satisfies both the tube's behavior and ohm's law.

A quiescent operating point (the Q point, the point where the circuit sits with no signal applied) is picked in the middle of that line. Picking the point somewhere in the middle allows enough room for the signal to swing both ways without the earth exploding. This point is picked by placing a resistor between the cathode and ground – the current flowing through it makes the cathode sit slightly above ground, which from the tube's perspective is the same as G1 being slightly negative.

The tube's maximum anode dissipation must also be taken into account so as to not fry the tube during operation - also known as $P_a \text{ max}$: $V_a \times I_a = 2.5W$ (for the EF80), this is shown as a curve on the same

graph. The load line must sit below this curve entirely otherwise the tube will start cooking, creating a hard constraint on where the Q point can be.

5. Gain and Bandwidth:

Input voltage swing on G1 produces a larger output voltage swing at the anode - the ratio between these two is called gain.

Bandwidth is the tube's maximum operating frequency before it starts noticeably losing gain.

The limiting factor when it comes to bandwidth is the tube's internal capacitance on top of the one added by connections and loose wiring - call it CL.

CL and RL form a low-pass filter which is meaningless at lower frequencies but at video frequencies it can become detrimental as the capacitor starts conducting more and more, effectively stealing current from the output. The frequency where gain has dropped considerably is the bandwidth limit.

Making RL bigger improves gain at the cost of bandwidth.

Gain and bandwidth basically fight each other, their product (gain $\times$ bandwidth) is fixed by gm and CL - that product cannot be changed, you can only trade one for the other.

6. Peaking Inductors:

As previously mentioned, RL and CL form a low-pass filter – the higher the frequency, the more CL conducts, the more output drops.

When an inductor is added in series with RL between the supply and anode, it will oppose the flow in current that CL is demanding. This results in a flatter gain as the upper limit of bandwidth is approached before roll off. This approach is called shunt peaking.

There is another trick called series peaking – a second inductor is added between the anode and the CRT cathode, splitting CL into two smaller capacitances with an inductor between them, forming a resonant LC network further expanding bandwidth.

Those two combined can extend bandwidth by 2 to 2.5 times compared to no peaking at all. However, the tradeoff of peaking inductors is that overdoing them causes the response to overshoot sharply leading to a bump instead of a flat top, which in turn causes ringing on sharp edges in the signal. Sync pulses and fast transitions would smear and overshoot.

7. DC Restoration:

The video signal contains both AC and DC. Passing this signal through any capacitor makes it AC-coupled, meaning the DC component of the signal, the one containing all the information regarding the black/blank and peak white levels, is now gone. The amplifier doesn't know what a black/blank is, nor does it care, this makes brightness completely random as the waveform is now free to drift around however it pleases with the black/blank and peak white levels turning into an average voltage on an AC signal instead of fixed points.

The solution to this is called DC restoration and can be achieved via a capacitor and diode together with a resistor that clamp the sync tips to a known reference voltage every single line:

- a) On first power-up, during active video, the diode is reverse biased, the capacitor remains discharged and nothing happens.
- b) When the polarity flips during the sync tips, the diode becomes forward biased and the capacitor charges.
- c) The polarity flips again, making the diode reverse biased. The capacitor now slowly discharges through the resistor with a time constant somewhere in the 10-100mS range. High enough to hold charge between sync tips, low enough to be able to track brightness changes over several frames. The sync tips are pinned by applying a DC bias to the other end of the capacitor, and conveniently enough, it can be the same capacitor that does the AC-coupling to begin with.

Think of it like a bucket (C) with water (charge) that has a hole (R) in it. The bucket must stay mostly full in between refills (sync pulses) but be small enough that if you pour less water in (dark scene), the bucket can drain down to the new level within the timespan of a few frames rather than staying full and making the image look artificially bright.

8. Grid Stopper Resistor:

A small resistor (in the range of hundreds of ohms) is placed right on the G1 pin on the tube socket. Its purpose is to stop the grid input capacitance from resonating with the lead inductance of the wiring, in turn causing the stage to oscillate at random VHF frequencies that are completely unrelated to the video signal.

9. Signal Path End to End:

Composite video is 1V p-p. Following the AC coupling the signal is fed to the G1 grid. This is also where the DC restoration clamp sits. Anode current is controlled by G1 voltage. As the video signal swings from 0V (sync tip, beam off) up to 1000mV (full picture brightness) the anode current follows accordingly. That current change across RL produces a voltage swing at the anode but inverted and larger.

The inverted, amplified signal drives the CRT cathode directly. High cathode voltage = beam suppressed = dark. Low cathode voltage = beam on = bright. The inversion is done by design. The peaking inductors sit in the anode-to-cathode path keeping the gain flat across the bandwidth's ceiling.

Before building our own let's take a look at the original amplifier:

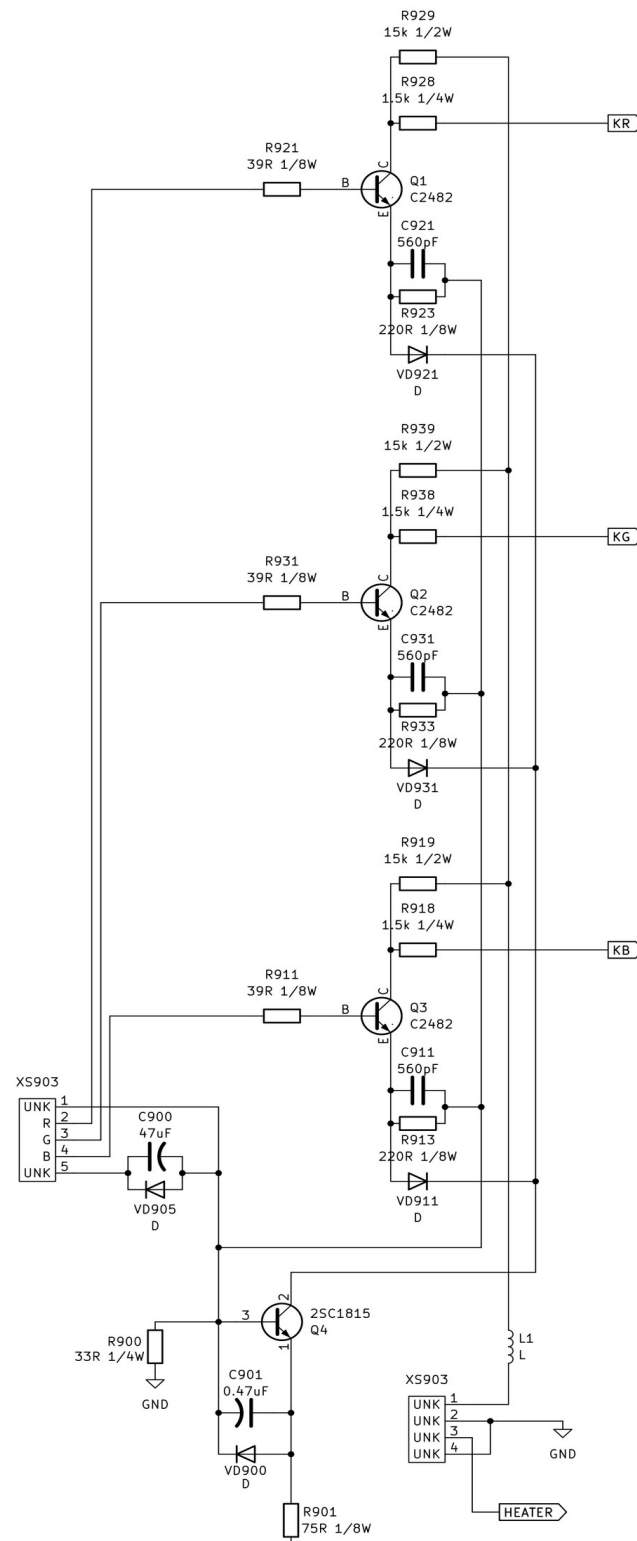


Figure 1.9 – Original video amplifier

I have kept the original numbering so as to keep track of the components more easily.

We can see the three identical BJT common-emitter stages using the 2SC2482, one for each color, all feeding each electron gun.

Behold, the bane of my existence:

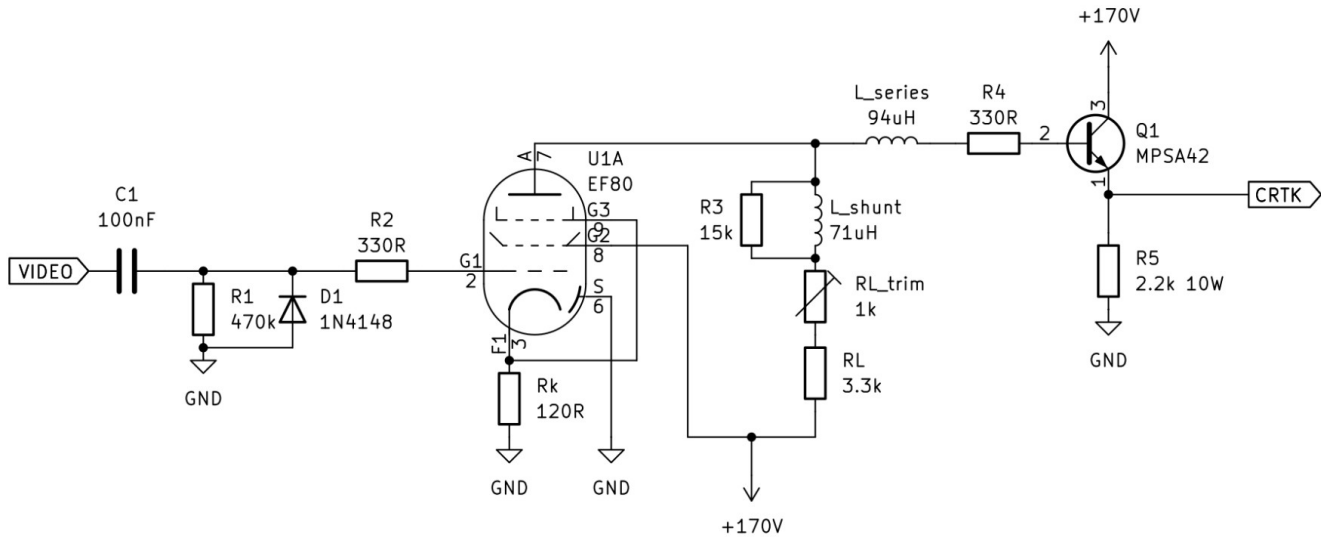


Figure 2.9 – The complete video amplifier

It seems rather simple and for what it is I can't argue with that granted you've worked with tubes or video amplifiers before unlike me.

1. AC-coupling/DC restoration:

Capacitor C1 serves both as the AC-coupling cap and as the DC restoration cap. This is done not only because it is convenient but also to save on board size and weight as I plan on attaching the board directly to the electron gun in order to minimise the total capacitance and maximise bandwidth.

A standard value in the hundred kilohm range is picked as well as a time constant and from there we calculate the capacitor's value:

$$\tau = R \times C$$

Rearranging for C:

$$C = \frac{\tau}{R}$$

Where R is 470k and τ is 50mS:

$$C = \frac{0.05}{470000} \approx 106 \text{ nF}$$

Nearest standard value is 100nF.

2. EF80 Pentode:

In order to calculate anything needed for the EF80 we must first pick a Q-point which is influenced by the supply voltage, screen (G2) voltage and the load line resistor:

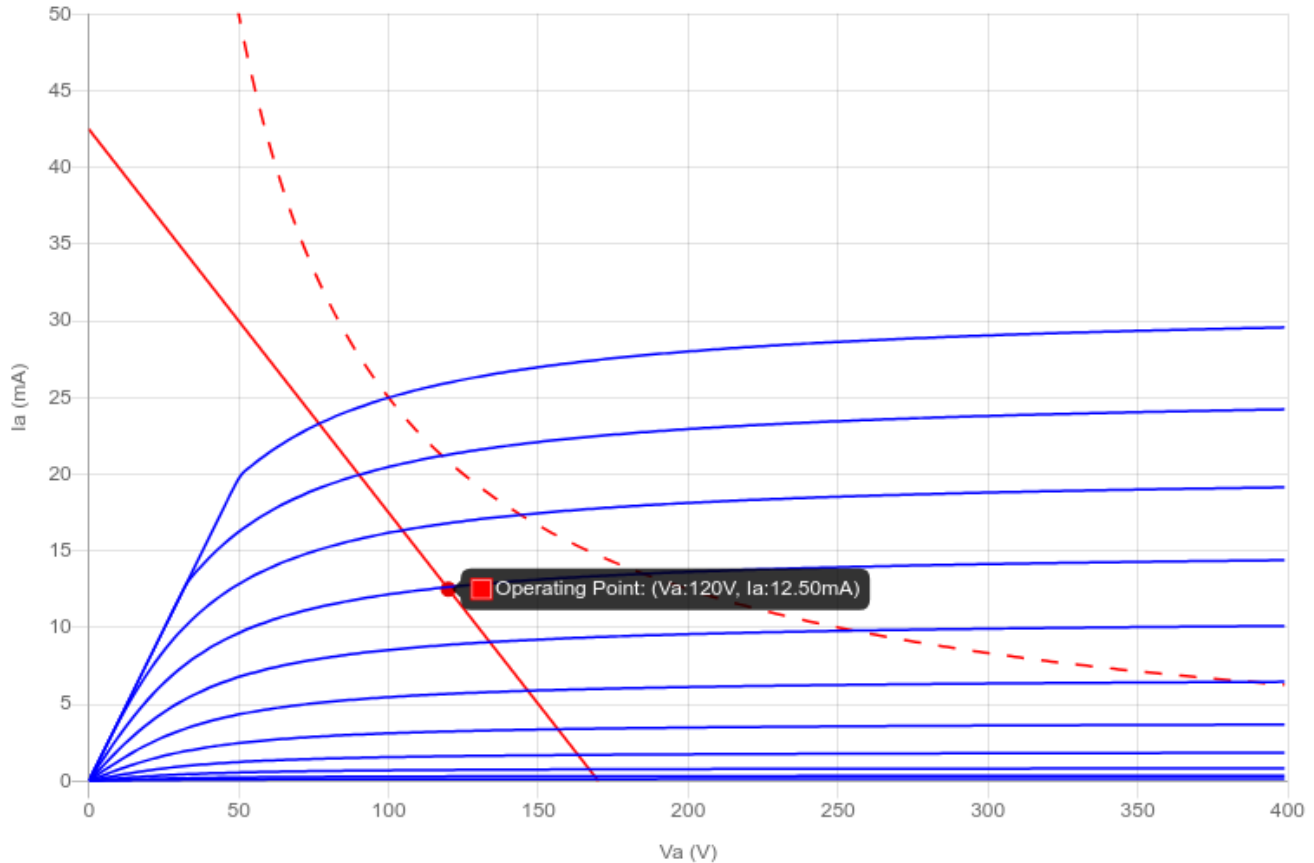


Figure 3.9 – Anode characteristic curves graph for the EF80

In order to find the Q-point on the graph, a supply voltage and a load line resistor value is picked first, with mine being 170V and 4k. We now need to identify the extremes of the line, with the first one being the maximum plate current:

$$I_{\max} = \frac{V_{cc}}{R_L}$$

Plugging in our values we get:

$$I_{\max} = \frac{170 \text{ V}}{4000 \Omega} = 42.5 \text{ mA}$$

We mark this extreme on the graph as well as the other one which is just the supply voltage and we draw a straight line. From here the only task left is picking a grid bias voltage (one of the blue lines), that is done based on the expected output swing and the application, mine comes out to -1.52V.

Another thing to keep in mind is that your line must sit firmly below the dotted line and not intersect with it by any means, otherwise your tube will become a really fancy lightbulb as that's the maximum plate dissipation rating of 2.5W.

Once that's done we're able to move onto verifying the plate load resistor (R_L) value, which not only sets the gain of the amplifier but also shifts the Q-point around. A lower value increases bandwidth at the expense of gain – this is caused by the changing current that goes through it causing a smaller voltage drop and thus a lower voltage swing, while also causing the anode plate to dissipate a larger amount of power as the resistor is dropping less voltage across itself resulting in a larger current, and vice versa. Its value can be checked using the following formula:

$$R_L = \frac{V_{HT} - V_a}{I_a}$$

Where V_{HT} is the supply voltage while V_a and I_a are the anode voltage and current picked off of the Q-point, which gives us:

$$R_L = \frac{170 - 120}{0.0125} = 4000 \Omega$$

I opted to use a fixed resistor of 3.3k in series with a trimmer of 1k as I do not know the required output swing. However you must take care so as to not exceed the power rating of the anode in either extreme by redoing the load line graph. Though to be noted that from here on out every formula and calculation follows the middle of that range (4k).

Before moving on we need to know the overall capacitance (CL) that we're dealing with, which unfortunately is partly estimated and comes from the EF80's anode (3.1pF), the base junction of the MPSA42 (3pF) and stray plus wire/trace capacitance (call it 5pF), that comes out to 11.1pF.

Now that we have CL we can calculate the gain-bandwidth product, the maximum frequency that this tube can switch at with a gain of one:

$$GBW = \frac{g_m}{2\pi \times C_L}$$

Transconductance (g_m or S) varies based on the supply voltage and is given by the datasheet to be 7.4mA/V at 170V V_{CC} , or 0.0074A. That gives us a gain-bandwidth product of:

$$GBW = \frac{0.0074}{2\pi \times 11.1 \text{ pF}} \approx 106 \text{ MHz}$$

Gain is given by $\frac{GBW}{f}$ where f is the targeted bandwidth, in our case, of 5MHz:

$$\text{Gain} = \frac{106}{5} = 21.2$$

That's not horrible per say but as I've mentioned previously, the required voltage swing on my CRT remains unknown, it might need a swing of 10V or 100V, so I plan on targeting a gain of 30, and if need be, limit it via the R_L trimmer, which is easily achievable with the help of peaking inductors:
Bandwidth without peaking inductors:

$$f_{BW} = \frac{1}{2\pi \times R_L \times C_L}$$

$$f_{BW} = \frac{1}{2\pi \times 4000 \times 11.1 \text{ pF}} \approx 3.58 \text{ MHz}$$

Bandwidth with shunt peaking only:

$$f_{BW} = \frac{1.41}{2\pi \times R_L \times C_L}$$

$$f_{BW} = \frac{1.41}{2\pi \times 4000 \times 11.1 \text{ pF}} \approx 5.05 \text{ MHz}$$

Bandwidth with series peaking only:

$$f_{BW} = \frac{2}{2\pi \times R_L \times C_L}$$

$$f_{BW} = \frac{2}{2\pi \times 4000 \times 11.1 \text{ pF}} \approx 7.16 \text{ MHz}$$

Bandwidth with both shunt and series peaking:

$$f_{BW} = \frac{2.5}{2\pi \times R_L \times C_L}$$

$$f_{BW} = \frac{2.5}{2\pi \times 4000 \times 11.1 \text{ pF}} \approx 8.96 \text{ MHz}$$

Moving on to the incredibly obvious choice:

For shunt peaking:

$$L_{shunt} = 0.4 \times R_L^2 \times C_L$$

$$L_{shunt} = 0.4 \times 4000^2 \times 11.1 \text{ pF} = 71.04 \mu\text{H}$$

Nearest standard value is 75uH.

For series peaking:

$$L_{series} = 0.53 \times R_L^2 \times C_L$$

$$L_{series} = 0.53 \times 4000^2 \times 11.1 \text{ pF} = 94.128 \mu\text{H}$$

Close enough to 100uH.

Standard practice is to use adjustable inductors for these, which is what I plan on doing. Worth mentioning is the damper resistor mounted in parallel with the shunt peaking coil. Its purpose is to kill the oscillation between the coil and CL by lowering the quality factor of the resonant circuit, damping out (hence the name a third time) high-frequency oscillations that would otherwise cause ghosting or ringing on the CRT display whenever the video signal hits a sharp edge. This is not needed for the series peaking coil.

Next up is the cathode resistor and G3:

$$R_k = \frac{|V_{g1}|}{I_a}$$

$$R_k = \frac{|-1.52\text{V}|}{0.0125} = 121.6 \Omega$$

Call it 120Ω.

As for G3, it simply needs to be connected to the cathode.

The only thing left is the grid stopper resistor, which is chosen empirically somewhere within 220-470Ω, I went with 330Ω as a starting point.

3. MPSA42 buffer stage:

If the EF80 stage were to be connected directly to the cathode, it would have to deal with its own capacitance alongside stray and wiring capacitance all on top of the CRT's capacitance, which could easily raise CL up to ~30pF and drop the bandwidth down to ~3.3MHz (peaking inductors included).

The easiest solution would be a common collector/emitter follower stage in between the CRT and pentode, meaning the tube now only has to deal with less than 12pF, while the transistor sees at worst 15pF.

As a good rule of thumb, the base resistor should be the same as the grid stopper resistor, while the collector is simply wired directly to the 170V supply, but the emitter isn't as straightforward:

$$R_e \leq \frac{1}{2\pi \times f_{\max} \times C_K}$$

Where $f_{\max}$ is the target bandwidth and C_K is the cathode capacitance plus stray, call it 12pF.

Now throwing those in:

$$R_e \leq \frac{1}{2\pi \times 5\text{ MHz} \times 12\text{ pF}} = 2652\ \Omega$$

Any higher in value and we're losing bandwidth, though to be noted that the main limitation on the value of this resistor is actually thermal management:

$$P = \frac{V_{\text{idle}}^2}{R_e}$$

Where V_{idle} is the Q-point voltage (120V):

$$P = \frac{120^2}{2652\ \Omega} = 5.4\text{ W}$$

With 2.2k that comes out to 6.5W, call it 20W for headroom.

To be noted that with 1k the power dissipation comes out to 14.4W.

Notice how a circuit with less than half the components of the horizontal deflection took up nearly just as many pages? Sit on that for a minute.

10. System Integration

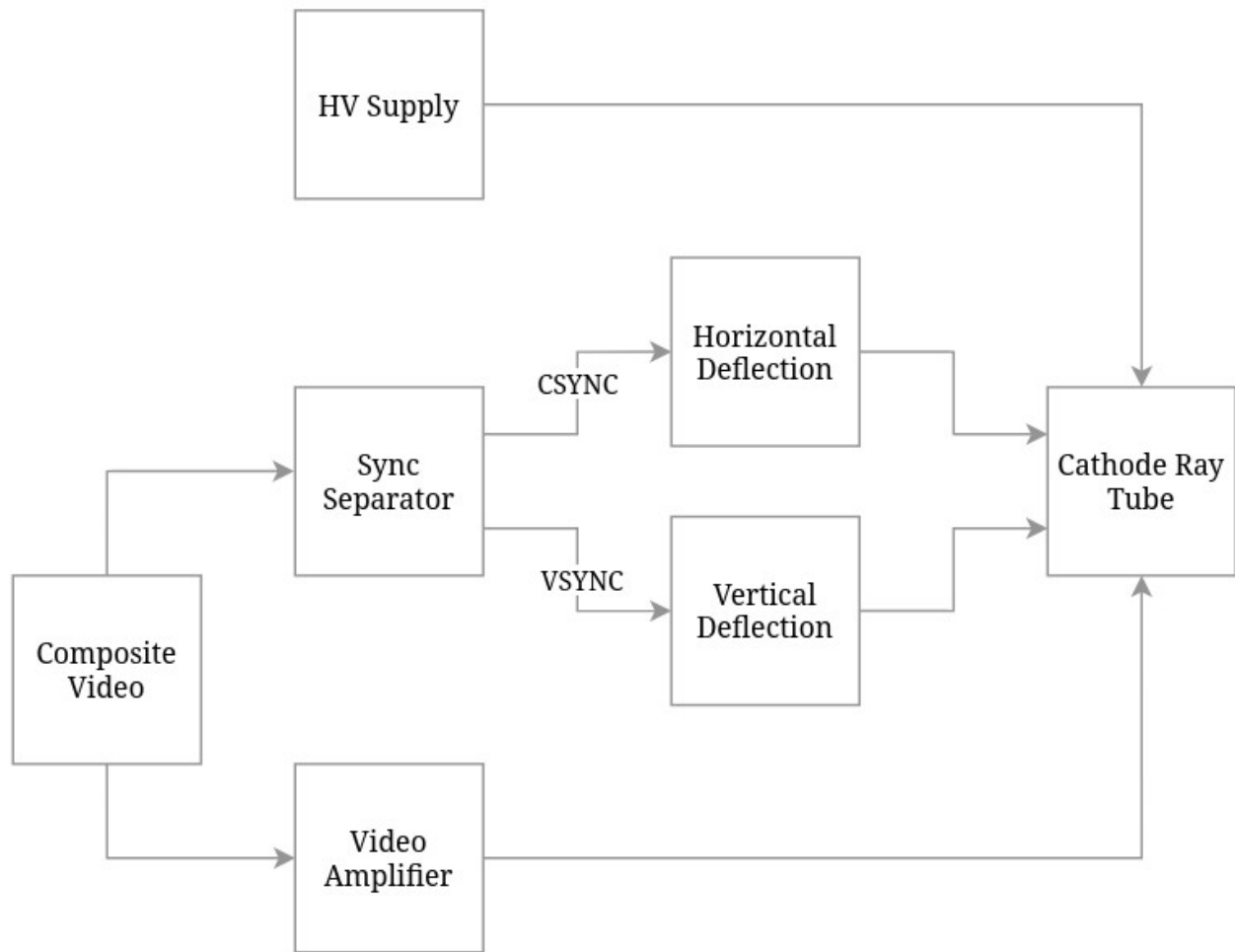


Figure 1.10 – The block diagram of the entire system

The incoming signal is referred to as "composite video" and is comprised of several different signals thrown onto the same wire which are the composite sync, the vertical sync, the picture information itself and the chroma subcarrier. We can't do anything with them like this so we feed the entire thing into the sync separator which, as the name implies, spits out the composite and vertical sync signals that are then fed into the oscillators of the horizontal and vertical deflections used to drive their corresponding yokes. At the same time, the entire composite video signal is fed directly to the video amplifier which connects to the cathode of the tube, with the HV supply running in the background accelerating the electrons into the phosphor screen.

12. Sources

1. Signal Spec:

[ITU-R BT.470-6 — PAL/NTSC signal specification](#)

2. Sync:

[LM1881 Video Sync Separator datasheet \(Rev. G\)](#)

3. HV Supply:

[Tesla Coil Safety Information](#)

Although meant for tesla coils, high voltage is high voltage.

[The Art of Electronics by Paul Horowitz & Winfield Hill - Third Edition](#)

Chapter 9.7, pages 652 through 656.

Note: I actually used the third edition *improved* but I couldn't find that one online.

[Transformer Characterisation Circuit – UZZORS2K's 555 Driver MKII](#)

[Ucx84x Current-Mode PWM Controllers datasheet \(Rev. H\)](#)

[UC3842/3/4/5 Provides Low-Cost Current-Mode Control](#)

[IRFP460 datasheet](#)

[Flyback Converter Snubber Design by Dr. Ray Ridley](#)

[Design Guidelines for RCD Snubber of Flyback Converters](#)

4. Horizontal Deflection:

[TV and Monitor Deflection Systems by Samuel M. Goldwasser](#)

Generally anything that has “Samuel M. Goldwasser” written somewhere on it is a gold mine.

[CD4046B Types datasheet](#)

[CD4046B Phase-Locked Loop: A Versatile Building Block for Micropower Digital and Analog Applications](#)

[CD4046 Explained and Example Circuit \(Phase-Locked Loop\)](#)

[Using 4046-Type PLLs Successfully – Anders Bøcher, December 2017](#)

[TLV237x 500- \$\mu\$ A/Ch, 3-MHz Rail-to-Rail Input and Output Operational Amplifiers With Shutdown](#)

[TIP31C datasheet](#)

[Ferrites and Accessories – TDK](#)

[FJL6920TU datasheet](#)

[VS-HFA08TB120-M3 datasheet](#)

5. Vertical Deflection:

[TDA1170D datasheet](#)

6. Video Amplifier:

[Universal Loadline Calculator for Vacuum Tubes](#)

[EF80 Pentode datasheet](#)

[Photofact Television Course – Howard W. Sams & Co., Inc.](#)

Pages 183 through 197.

[AN-1603: DC Restoration Circuit for an AC-Coupled Video Driver](#)

[Video Amplifiers](#)

7. Worth reading:

[The Electronics of Image Transmission in Color and Monochrome by V. K. Zworykin & G. A. Morton](#)